



RKDF UNIVERSITY, BHOPAL

B.Sc.(Biotechnology)

SCHEME

Semester – I

No	Subject Code	Subject Type	Subject Title	Marks Allotted						
				Assignment Marks		Theory Marks		Practical Marks		Total Marks
				Max	Min	Max	Min	Max	Min	
1	FC-101/1	Core	English Language	10	4	40	14	-	-	50
2	FC-101/2	Core	Development of Entrepreneurship	10	4	40	14	-	-	50
3	BBT 111	Core	Cell And Molecular Biology & Molecular Genetics	20	8	80	27	50	17	150
4	BCH 111	Core	Based on Inorganic, Organic And Physical Chemistry	20	8	80	27	50	17	150
5	BSB 111	Optional	Diversity of Microbes & Cryptogams	20	8	80	27	50	17	150
	BSZ 111	Optional	Invertebrates & Chordate	20	8	80	27	50	17	150
Total				80	32	320	109	150	51	550



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B.Sc.(Biotechnology)

B.C.A./B.A./ B.Com./B.Sc
Semester – I

Course	Subject	Subject Code
B.Sc. (Biotechnology)	English Language	FC-101/1

Unit-I

1. Amalkanti : Nirendranath Chakrabarti
2. Sita : Toru Dutt
3. Tryst with Destiny : Jawaharlal Nehru
4. Delhi in 1857 : Mirza Ghalib
5. Preface to the Mahabharata : C. Rajagopalachari
6. Where the Mind is Without Fear : Rabindranath Tagore
7. A Song of Kabir : Translated by Tagore
8. Satyagraha : M.K. Gandhi
9. Toasted English : R. K. Narayan
10. The Portrait of a Lady : Khushwant Singh
11. Discovering Babasaheb : Ashok Mahadevan

Unit-II

Comprehension

Unit-III

Composition and Paragraph Writing (Based on expansion of an idea).

Unit-IV

Basic Language Skills : Vocabulary – Synonyms, Antonyms, Word Formation, Prefixes and Suffixes, Words likely to be confused and Misused, Words similar in Meaning or Form, Distinction between Similar Expressions, Speech Skills

Unit-V

Basic Language Skills: Grammar and usage – The Tense Forms, Propositions, Determiners and Countable/Uncountable Nouns, Verb, Articles, Adverbs.

Prescribed Books: English Language and Indian Culture, Published by M.P. Hindi Grant Academy.

Note :- Eight questions to be set from unit-1 and four to be attempted.

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Semester – I

Course	Subject	Subject Code
B.Sc. (Biotechnology)	Development of Entrepreneurship	FC-101/2

Unit - I

Entrepreneurship- Definition, Characteristics and importance, Types and functions of an entrepreneur, merits of a good entrepreneur motivational factors of entrepreneurship.

उद्यमिता-परिभाषा, विशेषताएँ एवं महत्व, एक उद्यमी के प्रकार ,o dk;] , d vPn| m|eh d| x|.k] m|ferk vfhkij|.kk
?kVdA

Unit - II

- (a) Motivation to achieve targets and establishment of ideas. Setting targets and facing challenges. Resolving problems and creativity. Sequenced planning and guiding capacity, Development of self confidence.
- (b) Communication skills, Capacity to influence, leadership.
- v- y{; ikflr dh ij.kk ,o fopkj; dh LFkkiukA y{; fu/kkj.k ,o pukrh dk lkeukA
leL;k lek/kku ,o l' tukRedr;k क्रमबद्ध योजना एवं क्षमता की दिशाबद्धता।
आत्मविश्वास का विद्वान्
- c- lEi"t.k.k dykA iHkkfor dju dh {kerkA urRo

Unit - III

- (a) Project Report - Evaluation of selected process. Detailed project report – Preparation of main part of project report pointing out necessary and viability.
 - (b) Selecting the form of Organization – Meaning and characteristics of sole Proprietorship, Partnership and cooperative committees, elements affecting selection of a form of an organisation.
 - (c) Economic management – Role of banks and financial institutions banking, financial plans, working capital-evaluation and management, keeping of accounts.
- v- ifj;ktuk ifronu
puh gb: ifØ;k dk eY;kdu
विस्तृत परियोजना प्रतिवेदन— आवश्यकता एवं प्रासंगिकता परियोजना प्रपत्र के प्रमुख भाग परियोजना प्रतिवेदन
rl;kj djukA
- c- lxBu d: idkj dk pयन—एकाकी व्यवसाय, साझेदारी एवं सहकारी समिति का अर्थ एवं विशेषताएं संगठन के
p;u dk iHkkfor dju: oky: ?kVdA
- l- vkfFkd i;c/ku



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fofRr; I LfFku ,o cidk dh Hkfedk] cifdx] fofRr; ;ktuk] dk;dkjh iit h&eY; kdu rFk ic/ku] ykx r o eY; fu/kkj.k rFk ykHk dk eY; kdu] vkfFkd y:[kk&t k[kk j[kukA

Unit - IV

- (a) Production management. Methods of purchase. Management of movable assets/goods. Quality management. Employee management. Packaging.
- (b) Marketing management. Sales and the art of selling. Understanding the market and market policy. Consumer management. Time management.
- v- mRiknu dk ic/ku] [kjnhu d rjhd] py lEifRr@eky dk ic/ku] x.koRrk ic/ku] depkj h ic/ku] ifdx
- c- foi.ku ic/ku] fcØh ,o cpui dh dyk] cktkj dh le> ,o foi.ku uhfr] miHkkDrk ic/ku] le; ic/ku

Unit - V

- (a) Role of regulatory institutions – district industry centre, pollution control board, food and drug administration, special study of electricity development and municipal corporation.
- (b) Role of development organizations, khadi & village Commission/ Board, MP Finance Corporation, scheduled banks, MP Women's Economics Development Corporation.
- (c) Self-employment-oriented schemes, Prime Minister's Employment schemes, Golden Jubilee Urban environment scheme, Rani Durgavati Self-Employment scheme, Pt. Deendayal Self-employment scheme.
- (d) Various grant schemes – Cost-of-Capital grant, interest grant, exemption from entry tax, project report, reimbursement grant, etc.
- (e) Special incentives for women entrepreneurs, prospects & possibilities.
- (f) Schemes of M.P. Tribal Finance Development Corporation, schemes of M.P. Antyavasai Corporation, schemes of M.P. Backward Class and Minorities Finance Development Corporation.
- 1- fu;ked I LfFkvk dh Hkfedk&ftyk m।ग केन्द्र, प्रदूषण निवारण मंडल, खाद्य एवं औषधि प्रशासन, विद्युत विभाग तथा नगर निगम का विशेष अध्ययन।
- 2- विकासात्मक संस्थाओं की भूमिका, खादी एवं ग्रामीण आयोग/बोर्ड, मध्य प्रदेश वित्त निगम, अनुसूचित बैंक, मध्य प्रदेश का महिला आर्थिक विकास निगम।
- 3- Lojktxkj eyd ;ktuk, l & i/kkue=h jktxkj ;ktuk] Lo.k t;rh 'kgjh jktxkj ;ktuk] jkuh nxkorh Lojktxkj ;ktuk] fnun;ky Lojktxkj ;ktukA
- 4- विभिन्न अनुदान योजनाएँ— लागत पूँजी अनुदान, ब्याज अनुदान, प्रवेश कर से छूट, परियोजना प्रतिवेदन, प्रतिपूर्ति vunku vkfnA
- 5- महिला उद्यमियों हेतु विशेष प्रेरणाएँ, संभावनाएँ एवं समस्याएँ।
- 6- मध्य प्रदेश आदिवासी वित्त विकास निगम की योजनाएँ , m-i: vUR;kolk;h fuxe dh ;ktuk] e-i: fiNMk ox; ,o vYi l; ;d foRr fodkl fuxe dh ;ktuk, lA



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B.Sc.(Biotechnology)

Syllabus Semester – I

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	CELL AND MOLECULAR BIOLOGY & MOLECULAR GENETICS	BBT-111

UNIT-I

Cell as a basic unit - classification of cell types - cell theory - organization of plant and animals cells - comparison of microbial, plant and animal cells.

Ultra structure of cells - sub-cellular organization - structure and function of cell membranes, cytosol / Endoplasmic reticulum, nucleus, cytoskeleton, ribosomes, mitochondria and chloroplast, vacuoles, peroxisomes, lysosome, cell wall.

UNIT-II

Cell division (Eukaryotic and Prokaryotic) - mitosis, meiosis and cell cycle.

Cell - Chemical nature and macromolecular protein structure and function; membrane architecture, membrane associated process, ATP synthesis and photosynthesis, Sub-cellular organelles - mitochondria, chloroplast.

UNIT-III

Chromosomal Architecture - modern concept of Gene; Prokaryotes and Eukaryotes, Genetics of bacteria - Gene transfer in bacteria - Transformation, transduction and conjugation - mechanism and their use in genetic mapping, DNA transfections: Replication, repair, recombination, restriction and modification.

UNIT-IV

Gene as a unit of mutation and recombination - DNA as a genetic material- different types; RNA as a genetic material - different types-genetic code; Mutation - molecular nature - physical and chemical mutagens and its applications.

UNIT-V

DNA replication - prokaryotic and eukaryotic DNA replication - Mechanisms of DNA replication - Enzymes and accessory proteins involved in DNA replication - Bacterial genetic system - transformation, conjugation and transduction.

Reference

1. Cell Biology, De Roberties & De Roberties, Blaze publishers & Distributors Pvt. Ltd., New Delhi, 2001.
2. Molecular cell Biology (III rd Edition), Harvey Lodish, David Baltimore et al., W.H. Freeman, 2000.
3. Molecular Biotechnology - Principles and Applications of recombinant DNA, Glick, B.R and J.J Pasternak, 2002. Panima Publishing Co-operations.



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4. Microbial Genetics (II Edition) - Maloy S.R., Cronan, J.E.Jr and Friefelder, D. Jones. Bartlett publishers
5. Concept of Genetics (IV Edition) Willam S. Klug & Micheal R. Cummings. Molecular Biology, H.D.K.Kumar, Vikas Publishing House Private LTD.
6. 2. Introduction to Sub molecular Biology, Alberts Szent- Gyorgyi, 1960. Academic press.
7. The Molecular Biology of the Gene, J.D.Watson et al., 1987. Benjamin Cummings.
8. Genes V II. Benjamin Lewin, 1994. Oxford University Press. Oxford.
9. Molecular cell Biology (III rd Edition), Harvey Lodish, David Baltimore et al., W.H. Freeman, 2000.

PRACTICLE-I

Cell and Molecular biology & Molecular Genetics

1. Cell Types - Microbial, Animal and Plant Morphometric measurements.
2. Fractionation of cellular components.
3. Cell membrane, Separation and analysis of membrane components.
4. Mitosis and Meiosis
5. Microscopy and calibrations
6. Single Cell Colony Isolation - Checking for Genetic Markers
7. Induced Mutagenesis (UV and NTG)
8. Isolation of antibiotic resistant and auxotrophic mutants.
9. Cell counting methods :
10. Haemocytometer : WBS, RBC
11. Differential counting using Leishmans
12. Calibration using ocular micrometer
13. Finding out average cell size
14. Isolation of DNA from suitable materials (*E.coli*. Plant materials)
15. Microscopes and its parts
16. Mounting buccal epithelium and observing living cells using vital staining.
17. Mitosis in Onion root tip squash
18. Meiosis in grasshopper testis squash
19. Chironomous - Salivary gland Chromosome squash preparation
20. Staining of macro molecules- Carbohydrates and Lipids..
21. Observation o slides (Cardiac muscle, Sperm cell, Muscle cell).
22. Staining of Mitochondria in Yeast using Janus green.



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B.Sc.(Biotechnology)

Syllabus Semester – I

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	BASED ON INORGANIC, ORGANIC AND PHYSICAL CHEMISTRY	BCH-111

UNIT-I

Covalent Bond

Valence bond theory and its limitations, directional characteristics of covalent bond, various types of hybridization and shapes of simple inorganic molecules and ions (BeF₂, BF₃, CH₄, PF₅, SF₆, IF₇ SO₄, ClO₄)Valence shell electron pair repulsion (VSEPR)theory to NH₃, H₃O⁺, SF₄, ClF₃, ICl₂⁻ and H₂O. MO theory of hetero -nuclear (CO and NO) diatomic molecules, bond strength and bond energy.

UNIT-II

Gaseous States

Maxwell's distribution of velocities and energies (derivation excluded) Calculation of root mean square velocity, average velocity and most probable velocity. Collision diameter, collision number, collision frequency and mean free path. Deviation of Real gases from ideal behavior. Derivation of Vander Waal's Equation of State.

UNIT-III

Critical Phenomenon:

Critical temperature, Critical pressure, Critical volume and their determination. PV isotherms of real gases,.

Liquid States -Structure of liquids. Properties of liquids – surface tension, viscosity vapour pressure and optical rotations and their determination.

Solid State

Classification of solids, Laws of crystallography – (i) Law of constancy of interfacial angles (ii) Law of rationality of indices(iii) Law of symmetry. Definition of unit cell & space lattice. X ray diffraction by crystals. Derivation of Bragg equation. Liquid crystals :,types of liquid crystals. Applications of liquid crystals.

UNIT-IV

Stereochemistry of Organic Compounds-I

Concept of isomerism. Types of isomerism. elements of symmetry, molecular chirality,enantiomers, stereogenic centre, , chiral and achiral molecules with two stereogeniccentres, diastereomers, mesocompounds, resolution of enantiomers, inversion, retention and racemization.

Stereochemistry of Organic Compounds-II

Sequence rules, R & S systems of nomenclature. Geometric isomerism determination of configuration of geometric isomers. E & Z system of nomenclature, isomerism conformational



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analysis of ethane and n-butane, conformations of cyclohexane, Newman projection and Sawhorse formulae.

UNIT- V

Mechanism of Organic Reactions

Curved arrow notation, drawing electron movements with arrows, half-headed and double-headed arrows, homolytic and heterolytic bond breaking. Types of reagents – electrophiles and nucleophiles. Types of organic reactions. Energy considerations. Reactive intermediates carbocations, carbanions, free radicals, carbenes, arynes and nitrenes (formation, structure & stability). Assigning formal charges on intermediates and other ionic species.

Books:

1. Inorganic Chemistry, Huhee, Keiter and Keiter, Addition-Wesley, 1993.
2. Applied Mathematics for Physical Chemistry, J.R. Barante, Prentice Hall.
3. I. L. Levine, Quantum Chemistry, Prentice-Hall Inc., New Jersey.
4. T. Engel and P. Reid, Physical Chemistry, Benjamin-Cummings.
5. F. A. Carey and R. J Sundberg, Advanced Organic Chemistry, Part A and B, Plenum
6. P. S. Kalsi., Organic Reactions and their Mechanisms, New Age International

Paper (Practical's)

Section-A (Inorganic)

Volumetric Analysis

1. **Redox titrations:** Determination of Fe^{2+} , $\text{C}_2\text{O}_4^{2-}$ (using KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$)
2. **Iodometric titrations:** Determination of Cu^{2+} (using standard hyposolution).
3. **Complexometric titrations:** Determination of Mg^{2+} , Zn^{2+} by EDTA.

Section-B (Physical)

1. To determine the specific reaction rate of the hydrolysis of methyl acetate/ethyl acetate catalyzed by hydrogen ions at room temperature.
2. To prepare arsenious sulphide sol and compare the precipitating power of mono-, bi – and trivalent anions.

Section – C (Organic)

1. Preparation and purification through crystallization or distillation and ascertaining their purity through melting point or boiling point
2. (i) Iodoform from ethanol (or acetone)
(ii) *m*-Dinitrobenzene from nitrobenzene (use 1:2 conc. HNO_3 - H_2SO_4 mixture if fuming HNO_3 is not available)



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Syllabus Semester – I

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	Diversity of Microbes & Cryptogams	BSB 111

Unit-1

Viruses- Mycoplasma and Bacteria : characteristics of viruses and mycoplasma, general account of TMV and T4 bacteriophage. Bacterial structure, nutrition, reproduction and economic importance; general account of Cynobacteria.

Unit-2

Algae - General characters, classification and economic importance; important features and life history of Chlorophyceae- Volvox, Oedogonium, Charophyceae-Chara
Xanthophyceae - Vaucheria, Phaeophyceae - Ectocarpus, Sargassum, Rhodophyceae - Polysiphonia.

Unit-3

Fungi- general characters, classification and economic importance, important features and life history of Mastigomycotina- Phytophthora, Zygomycotina-Mucor. Asco mycotina -Aspergillus, Peziza, Basidiomycotina- Puccinia, Deuteromycotina- Cercospora, Colletotrichum, general account of Lichens.

Unit-4

Bryophyta - Classification, study of morphology, anatomy, reproduction of Hepaticopsida, Riccia, Marchantia, Anthrocerotopsida Anthoceros, Bryopsida- Polytrichum

Unit-5

Pteridophyta - Important characters and classification. Stelar organization. Morphology and anatomy of Rhynia. Structure, anatomy and reproduction in Lycopodium, Selaginella, Equisetum and Marsilea.

Suggested Books :

- 1.G.M. Smith 1971 Cryptogamic Botany. Vol - I Algae & Fungi Tata McGraw Hill Pub. Co. New Delhi.
- 2.G.M. Smith 1971 Cryptogamic Botany. Vol -II Bryophytes & Pteridophytes. Tata McGraw Hill Pub. Co. New Delhi.



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- 3.O.P.Sharma,1992. Text book of Thallophyta McGraw Hill Pub. Co.
- 4.O.P.Sharma,1990. Text book of Pteridophyta McMillan India Ltd .
- 5.P.D.Sharma 1991. The Fungi. rastogi & Co. Meerut.
- 6.H.C. Dubey.1990. an introduction of Fungi.Vikas Pub. house pvt.ltd.
- 7.P.Puri 1980. Bryophyta Atma Ram & Sons, Delhi.
- 8.A.Clifton.1958. Introduction to the Bacteria. McGraw Hillpub. Co.New delhi.

Practical Work : Semester-I

Scheme of practical examination

Marks:

Algae / Fungi	05	Brophyta	10	Pteridophyta
10				
Plant disease	05	Spotting	10	Sessional
				10
Total	50			



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Syllabus Semester – I

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	INVERTEBRATA & CHORDATA	BSZ 111

Unit-1

A brief introduction and nomenclature - Level of Organization. Phylum Protozoa: General characters- Classifications. Type study-Paramecium-Structure and Reproduction. General Topic: Protozoan Diseases.

Unit -2

Phylum Porifera: General characters- Type study- Ascon- Cellular structure. Phylum Coelenterata: Classification- Type study- Aurelia- Structure and life history. General Topic: Canal system in Sponges, Polymorphism in Coelenterates. Phylum Mollusca: General characters- Type study – Unio (Lamellidens) - External morphology and digestive system. General Topic: Mouth parts of Insects- Economic Importance of Mollusca.

Unit -3

Phylum Platyhelminthes: General Characters- Classification- Type study- Liver fluke- Structure and Reproduction. Phylum Annelida: General characters- Type study- Nereis- External morphology and reproduction. General Topic: Helminth Parasites in Man.

Unit-4

Introduction- Type Study: Amphioxus- external characters, digestive, excretory, respiratory, and circulatory systems.

Class: Pisces, General characters – Type Study: Scoliodon-External characters, Digestive, Excretory, Respiratory and Circulatory Systems- Structure of Brain-Sense organs and Reproductive system. General Topic: Accessory respiratory organs in fishes.

Unit-5

Class : Amphibia :General characters and classification -Type Study : Frog –External characters, Digestive ,Respiratory, Circulatory and Reproductive systems - Structure of Brain.

Class: Reptelia: General characters- Type study –Calotes- External characters-Digestive, Respiratory, Circulatory and Reproductive systems- Structure of Brain. General Topic: Identification of Poisonous and Non- Poisonous snakes. Golden age of Reptiles.

References

1. Agarwal V.K (2000) Invertebrate Zoology- S.Chand Company.
2. Barnes R.D (1987) Invertebrate Zoology- Saunders College Publications.
3. Barrington E.J (1981) Invertebrate Structure and Function. ELBS Editions.
4. Ekambaranatha Iyer (1993) Manual of Zoology Volume I Invertebrata.



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5. Kotpal R.L (2003) Modern text book of Zoology- Rostogi Publications, Meerut.
6. Ekambaranatha Iyer (1993) Manual of Zoology Vol.II, Viswanathan (printers& publishers) Chennai.
7. Jordon, E.L & Verma, P.S. (2000) Chordate Zoology, S.Chand & Co, New Delhi.
8. Newman H.H., Chordata, McMillan publishers.

SEMESTER-I

PRACTICE-I

INVERTEBRATA AND CHORDATA

I. Major Practicals:

Cockroach-Nervous, digestive, Reproductive system

Prawn-Nervous system

II. Minor Practicals:

Prawn –Appendages

Mouth parts –Honey Bee, Mosquito, and Cockroach.

III. Spotters:

a) Classify giving reactions:

Entamoeba, Paramecium, Leucosolenia, Hyalonema, Aurelia, Obelia, Taenia, Ascaris, Earthworm, Nereis, Penaeus, Freshwater mussel, Starfish, Cockroach, Amphioxus, Salpa, Frog, Cobra, Pigeon, Rabbit.

b) Draw Labelled Sketch:

T.S. of Taenia, T.S. of Fasciola, Ephyra larva, Nauplius larva, Zoea larva, Quill feather, Frog – Pectoral girdle, Pigeon –Pelvic girdle.

c) Biological Significance:

Sponge –Gemmule, Physalia, Leech, Limulus, Bipinnaria, Ascidian tadpole larva, Ichthyophis, Peripatus.

d) Relate structure and function

Taenia –Scolex, Nereis – Parapodium, Penaeus –Pereopods, Star fish –Tube feet (ventral view), Echinoderms, Draco, Bat.

e) Comment on Respiratory /skeletal/ dentition of the following

Star fish, Synsacrum, Dentition of Rabbit and Dog.



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Semester – II

No	Subject Code	Subject Type	Subject Title	Marks Allotted						
				Assignment Marks		Theory Marks		Practical Marks		Total Marks
				Max	Min	Max	Min	Max	Min	
1	FC-201/1	Core	हिन्दी भाषा संरचना	10	4	40	14	-	-	50
2	FC-201/2	Core	Environment	10	4	40	14	-	-	50
3	BBT 121	Core	Biomolecules & Organic Mechanisms In Biology	20	8	80	27	50	17	150
4	BCH 121	Core	Based on Inorganic, Organic And Physical Chemistry	20	8	80	27	50	17	150
5	BSB 121	Optional	Cell Biology & Genetics	20	8	80	27	50	17	150
	BSZ 121	Optional	Cell Biology & Genetics	20	8	80	27	50	17	150
Total				80	32	320	109	150	51	550



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B.Sc.(Biotechnology)

B.C.A./B.A./ B.Com./B.Sc

Semester – II

Course	Subject	Subject Code
B.Sc. (Biotechnology)	हिन्दी भाषा संरचना	FC-201/1

इकाई –1

भारत वंदना (काव्य)	:	सूर्यकांत त्रिपाठी 'निराला'
जाग तुझकों दूर जाना	:	सुश्री महादेवी वर्मा
स्वतंत्रता पुकारती (काव्य)	:	जयशंकर 'प्रसाद'
हम अनिकेतन (काव्य)	:	बालकृष्ण शर्मा 'नवीन'
		भाषा की महत्ता और उसके विविध रूप
		भाषा-कौशल

इकाई –2

करुणा (निबंध)	:	आचार्य रामचन्द्र शुक्ल
समन्वय की प्रक्रिया (निबंध)	:	रामधारी सिंह 'दिनकर'
बिच्छी बुआ (कहानी)	:	डॉ. लक्ष्मण विष्ट 'बटरोही'
अनुवाद	:	परिभाषा प्रकार, महत्व, विशेषताएं
		हिन्दी की शब्द-संपदा
		परिभाषिक शब्दावली

इकाई-3

विलायत पहुँच ही गया (आत्मकथांश)	:	महात्मा गांधी
अफसर (व्यंग्य)	:	शरद जोशी
तीर्थयात्रा (कहानी)	:	डॉ. मिथिलेश कुमारी मिश्र
मकड़ी का जाला (व्यंग्य)	:	डॉ. रामप्रकाश सक्सेना
		वाक्य-संरचना: तत्सम, तद्भव देशज विदेशी

इकाई-4

अप्य दीपो भव (वक्तृत्व कला)	:	स्वामी श्रद्धानन्द
भारत का सामासिक व्यक्तित्व (प्रस्तावना)	:	जवाहरलाल नेहरू
पत्र मैसूर के महाराजा को (पत्र-लेखन)	:	स्वामी विवेकानन्द
बनी रहेगी किताबें (आलेख)	:	डॉ. सुनीता रानी घोष
		पत्र-लेखन : महत्व और उसके विविध रूप
		सड़क पर दौड़ते ईहा मृग (निबंध) डॉ. श्यामसुन्दर दुबे

इकाई-5

योग की शक्ति (डायरी)	:	डॉ. हरिवंशराय बच्चन
कोश के अखाड़े में कोई पहलवान नहीं उतरता (साक्षात्कार)	:	भाषाविद् डॉ. हरदेव बाहरी से प्रो. त्रिभुवननाथ शुक्ल
नीग्रो सैनिक से भेंट (यात्रा-संस्मरण)	:	डॉ. देवेन्द्र सत्यार्थी
यदि बा न होती तो शायद गांधी को यह ऊँचाई न मिलती (साक्षात्कार)	:	5 कथाकार गिरिराज किशोर से सत्येन्द्र शर्मा
सर-लेखन, भाव-पल्लवन साक्षात्कार प्रयोजन और कौशल निर्धारित पाठ्य पुस्तक	:	"हिन्दी भाषा संरचना" मध्यप्रदेश हिन्दी ग्रन्थ अकादमी द्वारा प्रकाशित



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Semester – II

Course	Subject	Subject Code
B.Sc. (Biotechnology)	Environment @ i ;koj .k	FC-201/2

Unit-I

Study of Environmental and ecology:

- (a) Definition and Importance. (b) Environmental Pollution and problems.
 (c) Public participation and Public awareness.
 पर्यावरण एवं पारिस्थितिकीय अध्ययन
 (क) परिभाषा एवं महत्व (ख) पर्यावरण प्रदूषण एवं समस्याएँ
 (ग) जनभागीदारी एवं जन जागरण

Unit-II

Environmental Pollution :

- (a) Air, water, noise, heat and nuclear pollution.
 (b) Causes, effect and prevention of pollution.
 (c) Disaster management – Flood, Earthquake, cyclones and landslides.
 पर्यावरणीय प्रदूषण
 (क) वायु, जल, ध्वनि, ताप एवं आणविक-प्रदूषण (ख) प्रदूषण के कारण, प्रभाव एवं रोकथाम
 (ग) आपदा प्रबंधन- बाढ़, भूकंप, चक्रवात एवं भूस्खलन

Unit-III

Environment and social problems :

- (a) Development – non-sustainable to Sustainable. (b) Energy problems of cities.
 (c) Water preservation – rain-water collection.
 पर्यावरण और सामाजिक समस्याएँ
 (क) अधारणीय से धारणीय विकास (ख) नगरों की ऊर्जा समस्या
 (ग) जल संरक्षण- वर्षा, जल-संग्रहण

Unit-IV

Role of mankind in conserving natural resources :

- (a) Food resources – World food problem (b) Energy resources – increasing demand for energy.
 (c) Land resources – Land as resources.
 प्राकृतिक संसाधनों के संरक्षण में मनुष्य की भूमिका
 (क) खाद्य-आहार संसाधन – विश्व आहार समस्या (ख) ऊर्जा संसाधन- ऊर्जा की बढ़ती मांग
 (ग) भूमि संसाधन- भूमि संसाधन के गुप में, भूमि अवनयन, मनुष्यकृत भूस्खलन

Unit-V

Environment conservation laws :

- (a) Conservation laws for air and water pollution.
 (b) Wildlife conservation laws.
 (c) Role of information technology in protecting environment & health.
 पर्यावरण संरक्षण कानून
 (क) वायु तथा जल प्रदूषण-संरक्षण कानून
 (ख) वन्य प्राणि संरक्षण कानून
 (ग) पर्यावरण तथा स्वास्थ्य रक्षा में सूचना प्रौद्योगिकी की भूमिका



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Syllabus Semester – II

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	BIOMOLECULES & ORGANIC MECHANISMS IN BIOLOGY	BBT-121

Unit -I

Biomolecules: configuration and conformation. Properties of water as biological solvent. Carbohydrates: Chemical structures, nature, properties, Classification and Importance in Biological Systems.

Amino acids: Classification, properties, structure, nature. Proteins: Classification, Structure and Function. Primary, Secondary, Tertiary and Quaternary Structure and their functional significance.

Unit - II

Enzymes: Classification, Characteristics, Factors affecting enzyme activity. Enzyme Kinetics, K_m & V_{max} , Activation and Inhibition of Enzymes. Non- Protein Enzymes, Application of Enzymes *in vitro* and *in vivo*.

Unit - III

Nucleic acids: Bases, nucleosides and nucleotides, DNA & RNA structure, DNA forms, RNA types. Watson and crick model. Nucleosome model of chromosome. Genome organization and Packaging of DNA

Unit-IV

Carbohydrate Metabolism – Aerobic & Anaerobic glycolysis, sequence of reactions in glycolysis, regulation in glycolysis, citric acid cycle, glycogenesis, glycogenolysis (sequence of reactions & regulation), Pentose-phosphate pathway (sequence of reactions & regulation), extraction of energy from food sources.

Unit-V

Amino acid Metabolism – Amino acid breakdown (amino acid deamination, Urea cycle, metabolic breakdown of individual amino acids – glucogenic & ketogenic amino acids), amino acids as biosynthetic precursors (haem biosynthesis & degradation, biosynthesis of epinephrine, dopamine, serotonin, GABA, histamine, glutathione); biosynthesis of essential & non-essential amino acids.

Suggested Readings

1. Principle of Biochemistry : Leinger , A. L.
2. Biochemistry (1995) Lubert Stryer
3. Text Book of Biochemistry (1997) Devlin , Thomas M.
4. Biochemistry (1993) Geoffery, Zubay
5. Basic Biological Chemistry : Mahler and Cordes



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6. Harper's Review of Physiological Chemistry(1993) Murray, R. K., Mayes, P. A. Gramner, D. K. and Rowell V. W.
7. Lehninger's Principle of Biochemistry D L Nelson & M M Cox 5th Edition
8. Biochemistry – Keshav Trehan Wiley Eastern Publications
9. Fundamentals of Biochemistry-J.L.Jain S.Chand and Company
10. Biochemistry- Prasaraanga, Bangalore University
11. Fundamental of Biochemistry – Dr.A.C.Deb
12. Textbook of Organic Chemistry (A Modern Approach)
13. The Biochemistry of Nucleic acid – Tenth Edition-Roger L.P.Adams, John T. Knowler and David P.Leader, Chapman and Hall Publications

PRACTICE

1. Determination of pH from unknown biological samples using pH paper and pH meter.
2. Measurement of energy by Bomb calorimeter.
3. Analysis of carbohydrates (sugars, starch), proteins, lipids, and amino acids in biological samples.
4. Assays of amylase, peroxidase, catalase from biological samples using Spectrophotometer.
5. Separation of plant pigments and amino acids by PC and TLC.
6. Demonstration of gel electrophoresis.
7. Fractionation of biological material into its various components by Centrifuge.
8. Qualitative analysis of Carbohydrates, Proteins and Lipids.
9. Quantitative estimation of Protein by Folin-Lowry unitary method.
10. Quantitative estimation of sugar by Nelson Somogy's unitary method.
11. Quantitative estimation of sugar by DNS method.
12. Analyzing the enzyme activity.
13. Study the effect of pH on enzyme activity.
14. Study the effect of temperature on enzyme activity.
15. Determination of urine urea nitrogen.
16. Isolation of serum albumin by salting out method.
17. Determination of serum albumin by Bromocresol green method.
18. Determination of serum bilirubin.
19. Blood group analysis
20. Determination of haemoglobin content by haemoglobinomete



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Syllabus Semester – II

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	Based on inorganic, organic and physical chemistry	BCH-121

UNIT-I

Hydrogen Bonding – Definition, Types, effects of hydrogen bonding on properties of substances, application Brief discussion of various types of Vander Waals Forces

Metallic Bond and Semiconductors

Metallic Bond- Brief introduction to metallic bond, band theory of metallic bond
Semiconductors- Introduction, types and applications.

UNIT-II

Rate of reaction, rate equation, factors influencing the rate of a reaction – concentration, temperature, pressure, solvent, light, catalyst. Order of a reaction, integrated rate expression for zero order, first order, second and third order reaction. Half life period of a reaction. Methods of determination of order of reaction.

Kinetics-II

Effect of temperature on the rate of reaction – Arrhenius equation. Theories of reaction rate – Simple collision theory for unimolecular and bimolecular collision. Transition state theory of Bimolecular reactions.

UNIT-III

Electrolytic conduction, factors affecting electrolytic, Arrhenius theory of ionization, Ostwald's Dilution Law. Debye-Huckel – Onsager's equation for strong electrolytes (elementary treatment only) Transport number, definition and determination by Hittorff's methods, (numerical included). **Electrochemistry-II** Kohlrausch's Law, calculation of molar ionic conductance and effect of viscosity. Application of Kohlrausch's Law in calculation of conductance of weak electrolytes at infinite dilution. Applications of conductivity, Definition of pH and pKa, Buffer solution, Buffer action, Henderson – Hazelequation, Buffer mechanism of buffer action.

UNIT-IV

Nomenclature of alkenes, mechanisms of dehydration of alcohols and dehydrohalogenation of alkyl halides. The Saytzeff rule, Hofmann elimination, physical properties and relative stabilities of alkenes. Chemical reactions of alkenes mechanisms involved in hydrogenation, electrophilic and free radical additions, Markownikoff's rule, hydroboration–oxidation, oxymercuration reduction, ozonolysis, hydration, hydroxylation and oxidation with KMnO_4 .

UNIT-V

IUPAC nomenclature of branched and unbranched alkanes, the alkyl group, classification of carbon atoms in alkanes. Isomerism in alkanes, sources, methods of formation (with special reference to Wurtz reaction, Kolbe reaction Cycloalkanes nomenclature, synthesis of



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cycloalkanes and their derivatives –photochemical (2+2) cycloaddition reactions, dehalogenation of β,β -dihalides, ,pyrolysis of calcium or barium salts of dicarboxylic acids, Baeyer's strain theory and its limitations.

Book suggested :

1. Advanced Inorganic Chemistry, F.A. Cotton and Wilkinson, John Wiley.
2. Chemistry of the Elements. N.N. Greenwood and A. Earnshaw, Pergamon.
3. J. March., Advanced Organic Chemistry: Reactions, Mechanisms and Structure, John Wiley.
4. P. S. Kalsi., Organic Reactions and their Mechanisms, New Age International
5. J. P. Lowe and K.Peterson, Quantum Chemistry Academic Press.

Paper (Practical's)

Section-A (Inorganic)

Paper Chromatography

Qualitative Analysis of the any one of the following Inorganic cations and anions by paper chromatography (Pb^{2+} , Cu^{2+} , Ca^{2+} , Ni^{2+} , Cl^- , Br^- , I^- and PO_4^{3-} and NO_3^-

Section-B (Physical)

1. To determine the surface tension of a given liquid by drop number method.
2. To determine the viscosity of a given liquid.
3. To determine the specific refractivity of a given liquid.

SECTION – C (Organic)

1. Dibenzalacetone from acetone and benzaldehyde
2. Aspirin from salicylic acid. To study the process of) sublimation of camphor and phthalic acid

Books for practical:

1. Vogel's Textbook of Quantitative Analysis, revised, J. Bassett, R.C. Denney, G.H. Jeffery and J. Mendham, ELBS.
2. Synthesis and Characterization of Inorganic Compounds, W.L. Jolly. Prentice Hall.
3. Experiments and Techniques in Organic Chemistry, D.P. Pasto, C. Johnson and M.Miller, Prentice Hall.
4. Macroscale and Microscale Organic Experiments, K.L. Williamson, D.C. Heal



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BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	Cell Biology & Genetics	BSB 121

Unit-1

The cell envelope: plasma membrane, bilayer lipid structure, function of the cell wall. Structure and function of cell organelles: Golgi bodies, ER, Peroxisome, Vacuole, Chloroplast and Mitochondrion.

Unit-2

Ultrastructure and function of nucleus: Nuclear membrane, Nucleolus, Extranuclear genome, Presence and functions of mitochondrial and plastid-DNA, Plasmids. chromosomal organization; morphology, centromere and telomere, special types of chromosome, Mitosis and Meiosis

Unit-3

Variations in chromosomes structure : Deletions, duplications, translocations, inversions; variation in chromosome number, aneuploidy, polyploidy, DNA the genetic material, DNA structure and replication, the nucleosome model, satellite and repetitive DNA.

Unit-4

Structure of gene: genetic code, transfer of genetic information; transcription, translation, protein synthesis, tRNA, and ribosomes. Regulation of gene expression in prokaryotes and eukaryotes.

Unit-5

Genetic inheritance: Mendelism; laws of segregation and independent assortment; linkage analysis; interactions of genes. Genetic variations; mutations, spontaneous and induced; transposable elements; DNA damage and repair.

Suggested Books :

1. Alberts B.D. Lewis, J. Raff, M. Roberts, K. and Watson I.D. 1999 molecular Biology of cell Garland Pub. Co. Inc. New York, U.S.A.
2. P.K. Gupta 1999 A text Book of cell and Molecular Biology, Rastogi Pub. Meerut India.
3. Kleinsmith L.J. and Molecular Biology (2nd edition) Harper Collins College pub. New York USA.
4. P.K. Gupta Genetic's Rastogi Pub. Meerut.
5. Sinha & Sinha cytogenetics & plant Breeding Vikas Pub.



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Syllabus Semester – II

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	CELL BIOLOGY & GENETICS	BSZ 121

UNIT 1

Prokaryotic and eukaryotic cells –Ultrastructure and Organization. Plasma membrane–Ultra structure–Chemical composition and functions of modifications of plasma membrane. Endoplasmic reticulum: Morphology, Ultra structure, chemical composition and functions. Golgi complex: Ultra structure, chemical composition and functions.

UNIT 2

Lysosomes: Ultra structure and polymorphism- chemical composition and functions: Peroxisomes and glyoxysomes. Mitochondria: Ultra structure- chemical composition-enzyme systems- functions-Oxidation- Respiratory chain (ETP)- Kreb's cycle, ATP Production and Biogenesis.

UNIT 3

Ribosomes: Ultra structure-types- chemical composition - functions. Nucleus and Neucleolus: Ultra structure of Nucleus and Nucleolus. Nucleic Acids: DNA –Ultra structure-replication-transcription, Chromosomes: Ultra structure of Chromosomes and Giant Chromosomes, Cell division- mitosis and meiosis. Cell cycle,

UNIT 4

Introduction –Laws of Mendel –Interaction of genes (Epistatic gene ,Complementary genes and Lethal genes . Inheritance of Blood group in man and Coat colour in Rabbit . Mechanism of linkage and crossing over –Types and theories - Chromosomal Mapping –Sex linked inheritance (haemophilia, and colour blindness). Sex limited inheritance and sex influenced inheritance .

UNIT 5

Sex determination in man, Drosophila and Bonellia. Mutations – Point mutation and Chromosomal aberrations and mutagens . Inbreeding and out breeding, heterosis –Genetic applications in animals. DNA as genetic material –Experiments . Syndromes (Down syndrome and Turners syndrome in man) .

TEXT BOOKS:

Cell biology. Veer Bala Rastogi, Rastogi Publications.

Cell Biology, Power.Verma P.S. and Agarwal V. K . –Concepts of Genetics .

Rastogi V.B. A text book of Genetics, K.Ramnath, Meerut.



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**YEAR-I, SEMESTER-II
PRACTICAL
CELL BIOLOGY AND GENETICS**

A. Major Practicals

Use Microscopes, Camera Lucida, Stage and Ocular micrometers.

Total Counting of RBC / WBC Using haemocytometer.

Blood Smear Preparation, Differential count of WBC.

Mounting Buccal Epithelium and observing living cells using vital staining.

Study of mitotic division using onion root tips.

Study of prepared slides of different tissues.

Submission of practical record.

B.GENETICS PRACTICALS

Observation of common mutants of drosophila

Preparation of mounting of the salivary gland in chironomous larva

Submission of practical record.



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SCHEME

Semester – III

No	Subject Code	Subject Type	Subject Title	Marks Allotted						
				Assignment Marks		Theory Marks		Practical Marks		Total Marks
				Max	Min	Max	Min	Max	Min	
1	FC-301/1	Core	Values & Spirituality	10	4	40	14	-	-	50
2	FC-301/2	Core	Environment	10	4	40	14	-	-	50
3	BBT 131	Core	Food Biotechnology & Bioprocess Engineering	20	8	80	27	50	17	150
4	BCH 131	Core	Based on Inorganic, Organic And Physical Chemistry	20	8	80	27	50	17	150
5	BSB 131	Optional	Diversity & systematics of seed plants	20	8	80	27	50	17	150
	BSZ 131	Optional	Animal Physiology & Developmental Biology And Immunology	20	8	80	27	50	17	150
Total				80	32	320	109	150	51	550



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Semester – III

Course	Subject	Subject Code
B.Sc. (Biotechnology)	Values & Spirituality	FC-301/1

Chapter 1: VALUE EDUCATION

1.1	Objectives	1.2	Introduction	1.3	Concepts of Values
1.4	Definition and Types of values			1.5	The need for Education in values
1.6	Challenges for Value adoption			1.7	Character development
1.8	Vision of a better world			1.9	Summary
1.10	Glossary			1. 11	Suggested reading

Chapter 2: INCULCATION OF VALUES

2.1	Objectives	2.2	Introduction	2.3	Classification of values
2.4	Personal Values	2.5	Family Values	2.6	Social Values
2.7	Spiritual values	2.8	Benefits of value adoption	2.9	Summary
2.10	Glossary	2.11	Suggested reading		

Chapter 3: MAJOR RELIGIONS OF THE WORLD

3.1	Objectives	3.2	Introduction	3.3	Hinduism
3.4	Jainism	3.5	Buddhism	3.6	Christianity
3.7	Islam	3.8	Sikhism	3.9	Summary
3.10	Glossary	3.11	Suggested reading		

Chapter 4: EXPLORING THE SELF

4.1	Objectives	4.2	Introduction	4.3	Anatomy of the self
4.4	The cyclic processes within the self			4.5	States of the awareness
4.6	Innate qualities	4.7	Acquired qualities	4.8	Empowering the self
4.9	Summary	4.10	Glossary	4.11	Suggested reading

Chapter 5: THOUGHT AND THE THINKER

5.1	Objectives	5.2	Introduction	5.3	Know the mind(Team)
5.4	Thought power	5.5	Types of thoughts	5.6	Thinking process
5.7	Positive thinking	5.8	Power and Acts	5.9	Summary
5.10	Glossary	5.11	Suggested reading		



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Semester – II

Course	Subject	Subject Code
B.Sc. (Biotechnology)	Environment	FC-301/2

Unit - I

Problems of natural resources:

- (a) Problems of water resources- Utilization of surface and ground water overutilization, flood, draught, conflict over water, Dams and related problems.
- (b) Problems of forest resources -uses and over utilization, deforestation, Dams and its effects on forests and tribes.
- (c) Problems of land resources -land as a source , erosion of land, man-induced, land slides and desertification.

प्राकृतिक संसाधन की समस्याएँ

- (क) जल संसाधन की समस्या— सतह एवं भूजल का उपयोग , अति दोहन बाढ़ सूखा , जल पर संघर्ष , बाँध —लाभ एवं समस्याएँ
- (ख) वन संसाधन की संस्याएँ —उपयोग एवं अति दोहन , वनोन्मूलन , इमारती लकड़ी, निस्सारण , बाँध एवं उनका बन और आदिवासीयो पर प्रभाव
- (ग) भूमि संसाधन की समस्याएँ— स्त्रोत के क्रय में भूमि का अवभ्रमण , मानव प्रेरित भू — स्खलन और मरुस्थीकरण

Unit - II

Bio-diversity and its protection -

- (a) Value of bio-diversity -consumable use: productive use, social, alternative, moral asthetic and values
- (b) Bio - diversity and multi -diversity at global and national levels.
- (c) Threats to bio -diversity -loss of habitat, poaching of wildlife, man wildlife conflicts.

जैव विविधता और उसका संरक्षण –

- (क) जैव विविधता का मूल्य – उपभोग्य , उपयोग , उत्पादक उपयोग, सामाजिक, नैतिक सौर्दयंगत तथा वैकल्पिक मूल्य
- (ख) वैश्विक , राष्ट्रीय तथा स्थानीय स्तरो पर जैव विविधता वृहत विविधताओ के राष्ट्र रूप में भारत ।
- (ग) जैव विविधता के खतरे – आवासीय हानि ,वन्य जीवन में अनाधिकार घुसपैठ तथा मानव , वन्य जीवन —संघर्ष

Unit - III

Human population and environment

- (a) Population growth, disparities between countries.
- (b) Population explosion, family welfare programme
- (c) Environment and human health

पर्यावरण और जनसंख्या तथा पर्यावरण

- (क) जनसंख्या – वृद्धि , राष्ट्रों के बीच अंतर
- (ख) जनसंख्या —विस्फोट , परिवार कल्याण कार्यक्रम
- (ग) पर्यावरण और मानव स्वास्थ्य



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Unit - IV

Multidisciplinary nature of environmental studies :

- (a) Natural resources
- (b) Social problems and the environment
- (c) Environmental awareness

पर्यावरण और उसका बहुअनुशासनिक स्वरूप

- (क) प्राकृतिक संसाधन
- (ख) सामाजिक समस्याएँ और पर्यावरण
- (ग) पारिस्थिति तन्त्र

Unit - V

Multidisciplinary nature of environmental studies :

- (a) Natural resources
- (b) Social problems and the environment
- (c) Environmental awareness

पर्यावरण और उसका बहुअनुशासनिक स्वरूप

- (क) प्राकृतिक संसाधन
- (ख) सामाजिक समस्याएँ और पर्यावरण
- (ग) पारिस्थिति तन्त्र



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Syllabus Semester – III

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	FOOD BIOTECHNOLOGY & BIOPROCESS ENGINEERING	BBT-131

Unit-I

Food as a substrate for microorganisms: pH, moisture content, redox potential, nutrient content and inhibitory substances.

Microorganisms important in food industry: Molds, identification of molds of industrial importance, yeasts & yeasts like fungi, yeasts of industrial importance, bacteria.

Unit-II

General principles underlying spoilage: Causes of spoilage, factors affecting kinds and members of microorganisms in food, factors affecting growth of microorganisms in food. Chemical changes caused by microorganisms.

Unit-III

Foods and enzymes produced by microorganisms – Bread, malted beverages, wines, distilled liquors, vinegar, fermented vegetables, fermented dairy products & oriental fermented goods. Microorganism as food: single cell protein, fats and amino acids from microorganisms. Production of microbial enzymes.

Unit-IV

Introduction to fermentation: rate of microbial growth and death. Fermentation - types, classification, basic requirements, design of a fermentor, factors involved in fermentor design - basic functions - containment body construction - temperature control-stirring and mixing – viscosity

UNIT-V

Isolation and preservation of industrially important micro organisms - strain development mutation and recombination - upstream processing, media for industrial fermentation - characteristics of an ideal production medium - raw material - screening for production media - media formulation - sterilization - (batch and continuous) - addition of antifoaming agents.

References:

1. Fraizier, Food Microbiology, 1978, McGraw Publishers.
2. Pelczar: Microbiology.
3. Prescott: Microbiology.
4. Stanbury P.P. and Whitaker, A. 1984. Principles of Fermentation Technology. Pergamon Press, Oxford UK.
5. Steinkraus, K.H. 1983. Handbook of Indigenous Fermented Foods. Marcel Dekker, New York.



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6. Casida, J.E., 1968. Industrial Microbiology, Wiley Eastern Publication
7. Cruegar, W. and Cregar, A., 1989. Biotechnology: A text book of industrial Microbiology, 2nd edition. Panima Publishing Corporation, New Delhi.
8. Patel. A.H. 1966. Industrial Microbiology, Mac Millan India Ltd.
9. Stanbury, A.H., A. Whittaker and Hall S.J. 1995. Principles of fermentation technology 2nd edition, Pergamon Press.

YEAR-I SEMESTER-III

PRACTICE: FOOD BIOTECHNOLOGY & BIOPROCESS ENGINEERING

1. Quality testing of milk by resazuring test.
2. Determination of phasphatase activity in butter, whey, milk powder.
3. Microbiological analysis of food production.
4. Presumptive test for coliform in butter.
5. Analysis of mycotoxin in fungal contaminated food materials.
6. Introduction of fermenter.
7. Isolation of industrially important microorganisms.
8. Determination of thermal death point (TDP) and thermal death time (TDT) of microorganism for design of a sterilizer.
9. Suggested Readings
10. Plant tissue culture : Bajaj, Y.P.S. Series.
11. Plant tissue culture : Gamborg and Phillip.
12. Basic and Agricultural Biotechnology (1993) Purohit and Mathur
13. Plants, Genes, and Agriculture : Chrispeels, M. J. and Grierson, D.
14. Genetic Engineering of crop plants : Lycett, G. W. and Grierson, D.
15. Biotechnological innovation in Animal productivity: (Biotol Series)
16. Culture of Animal cell: A mannual of Basic Techniques(4th ed.) (2000) Freshney



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Syllabus Semester – III

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	Based on inorganic, and Physical Chemistry	BCH-131

UNIT-I

Definition of transition elements, position in the periodic table, General characteristics & properties of 1st transition elements, Structures & properties of some compounds of transition elements– TiO_2 , VOCl_2 , FeCl_3 , CuCl_2 and $\text{Ni}(\text{CO})_4$. **Chemistry of Elements of IInd & IIIrd transition series** General characteristics and properties of the IInd and IIIrd transition elements Comparison of properties of 3d elements with 4d & 5d elements with reference only to ionic radii, oxidation state, magnetic and Spectral properties and stereochemistry.

UNIT-II

Werner's coordination theory, effective atomic number concept, chelates, nomenclature of coordination compounds, isomerism in coordination compounds, valence bond theory of transition metal complexes. **Non-aqueous Solvents** Physical properties of a solvent, types of solvents and their general characteristics, reactions in non-aqueous solvents with reference to liquid NH_3 and liquid SO_2 .

UNIT-III

Definition of thermodynamic terms: system, surrounding etc. First law of thermodynamics: statement, definition of internal energy and enthalpy. Heat capacity, heat capacities at constant volume and pressure and their relationship. Joule's law – Joule – Thomson coefficient for ideal gas and real gas: and inversion temperature. **Thermodynamics-II** Calculation of w.q. dU & dH for the expansion of ideal gases under isothermal and adiabatic conditions for reversible process, Temperature dependence of enthalpy, Kirchhoff's equation. Bond energies and applications of bond energies.

UNIT-IV

Monohydric alcohols nomenclature, methods of formation by reduction of aldehydes, ketones, carboxylic acids and esters. Hydrogen bonding. Acidic nature. Reactions of alcohols. **Epoxides** Synthesis of epoxides. Acid and base-catalyzed ring opening of epoxides, orientation of epoxide ring opening.

UNIT-V

Nomenclature of Carboxylic acids, structure and bonding, physical properties, acidity of carboxylic acids, Preparation of carboxylic acids. Reactions of carboxylic acids. **Acid Derivatives** Structure, nomenclature and preparation of acid chlorides, esters, amides and acid anhydrides. Relative stability of acyl derivatives. Physical properties, inter conversion of acid derivatives by nucleophilic acyl substitution. Mechanisms of esterification and hydrolysis (acidic and basic).



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Book suggested :

1. R. G. Mortimer, Mathematics for Physical Chemistry Elsevier.
2. F. L. Pilar, Elementary Quantum Chemistry , Dover Publication
3. E. L. Eliel, Stereochemistry of Carbon Compounds, McGraw-Hill
4. S. M. Mukherji and S. P. Singh, Reaction Mechanism in Organic Chemistry,
5. Comprehensive Coordination Chemistry eds., G. Wilkinson, R.D. Gillars and J.A. McCleverty, Pergamon.

Paper (Practicals)

Section – I (Inorganic)

1. Gravimetric Analysis

Quantitative estimations of, Cu^{2+} as copper thiocyanate and Ni^{2+} as Ni – dimethylglyoxime.

Section-B (Physical)

1. To determine the CST of phenol – water system.
2. To determine the solubility of benzoic acid at various temperatures and to determine the ΔH of the dissolution process
3. To determine the enthalpy of neutralisation of a weak acid/weak base vs. strong base/strong acid and determine the enthalpy of ionisation of the weak acid/weak base.

Section-C (Organic)

Systematic identification (detection of extra elements, functional groups, determination of melting point or boiling point and preparation of at least one pure solid derivative) of the following simple mono and bifunctional organic compounds: Naphthalene , anthracene, acenaphthene, benzyl chloride, *p*-dichlorobenzene, *m*-dinitrobenzene, *p*-nitrotoluene, resorcinol , hydroquinone, 1-naphthol, 2-naphthol, benzophenone, ethyl methyl ketone, benzaldehyde, vanillin, oxalic acid, succinic acid,



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B.Sc.(Biotechnology)

Third Semester

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	Diversity & systematics of seed plants	BSB 131

UNIT – 1

Characteristics and Classification of Gymnosperms, Heterospory and Origin of Seed Habit, Evolution and Diversity of Gymnosperms, Geological Time Scale, and Fossilization. Fossil Gymnosperms: Lyginopteris and Lagenostoma.

UNIT – 2

Morphology, Anatomy Reproduction and life cycle of Cycas, Pinus and Ephedra.

UNIT – 3

Origin and Evolution of Angiosperms, Fundamental components of α , β , γ taxonomy, Plant Identification, Principles and rules of Botanical Nomenclature, Herbarium and Botanical gardens; Classification of Angiosperms: Bentham and Hooker, and Hutchinson, Modern trends in Taxonomy.

UNIT – 4

Diagnostic characteristics and Economic Importance of Families –Ranunculaceae, Brassicaceae, Malvaceae, Rutaceae, Fabaceae, and Apiaceae.

UNIT – 5

Diagnostic characteristics & Economic Importance of Families –Asteraceae, Asclepiadaceae, Solanaceae, Lamiaceae, Euphorbiaceae, Liliaceae and Poaceae.

Practical Exercises + Scheme (Marks- 50)

- Gymnosperms- 10
- Morphological and anatomical study of Cycas, Pinus, and Ephedra (all parts).
 - Study of permanent slides of Cycus, Pinus and Ephedra.
- Angiosperms- 15
- Study of types of inflorescence and flowers with labelled sketches.
 - Technical description of common flowering plants belonging to families mentioned in theory syllabus.
- Spotting- 10
- Viva- voce- 5
- Practical record - 10



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SUGGESTED READINGS:--

1. Agarwal, S.B. 2007. Unified Botany, Shivalal Agarwal & Company Indore.
2. Bhatnagar, S. P. and Moitra 1996. Gymnosperms. New Age International Limited, New Delhi.
3. Davis, P.H. and Heywood, V.H. 1963, Principles of Angiosperm taxonomy. Oliver and Boyd, London.
4. Gangulee, H. C. & Kar, A. K. 2006. College Botany Voll.III, New Central Book Agency (P) Ltd.
Kolkata, 700009.
5. Heywood, V.H. and Moore, D.M. (eds) 1984. Current concepts in plant taxonomy. Academic press
London.
6. Jeffery, C. 1982. An Introduction of plant taxonomy. Cambridge University Press Cambridge, London.
7. Jones, S.B. Jr. and Luchsinger, A.E. 1986. Plant Systematic. Mc Graw Hill Book Co. New York.
8. Kaushik, M.P. 2003. Modern Textbook of Botany, Prakash Publication Muzaffar Nagar U.P.
9. Mukherjee, S.K. 2006. College Botany Voll.II, New Central Book Agency (P) Ltd. Kolkata, 700009.
10. Pandey, B. P. 2010. A Text book of Botany- Angiosperms, S. Chand & Company Ltd. Ramnagar,
New Delhi- 110055.
11. Radford, A.E. 1986. Fundamentals of Plant Systmatics, Happer and Raw, New York.
12. Saxena and Sarabhai. 1989. Text book of Botany. Rastogi Publication Meerut.
13. Singh, G. 1999. Plant Systematics : Theory and Practice. Oxford and IBH Pvt. Ltd. New Delhi.
14. Vasishta, P.C. 2005. Botany for degree students Voll. V, Gymnosperms. S. Chand & Company Ltd.
Ramnagar, New Delhi- 110055.



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Syllabus Semester – III

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	ANIMAL PHYSIOLOGY & DEVELOPMENTAL BIOLOGY AND IMMUNOLOGY	BSZ 131

Unit-1

Nutrition –types, Enzymes – Enzyme action, Coenzymes, Digestion in man. Respiration – Respiratory pigments, role in transport of O₂ and CO₂ in man. Circulation - blood composition, origin and conduction of heart beat in an – blood pressure, Heart diseases– heart attack.

Unit-2

Excretion – types of nitrogenous wastes – structure of the mammalian kidney and urine formation – renal failure – kidney stone – kidney transplantation. Osmotic – ionic regulation in freshwater, marine, estuarine and terrestrial organisms

Unit-3

Amoeboid, ciliary and flagellar movements. Types of muscles – ultra structure of skeletal muscle – Muscle contraction and theories. Neuron, - Types- Impulse transmission, -synaptic transmission - reflex action.

-Endocrine glands in man, secretions and disorders.

Unit-4

Spermatogenesis – definition – process and significance, structure of mammalian sperm. Oogenesis – definition – process and significance – Types of eggs and egg membranes. Fertilization – definition – process and significance. Parthenogenesis – definition and significance – types of parthenogenesis.

Unit-5

Cleavage patterns (types) – Cleavage in Frog, Chick and Mammals. Morula and Blastulation.

Introduction – cells and organs involved in immune response. Types of immunity – Innate & adaptive immunity of acquired immunity, humoral and cell mediated immunity, active and passive immunity.

TEXT BOOKS:

Verma P.S. & Tyagi B.S. Animal Physiology, 6th edition. S.Chand & Co.

Agarwal, V.K. Agarwal, R.A.Srivastava A.K. & Kausha Kumar, Animal physiology & Biochemistry, S. Chand & Co.,

De Beer, G.R. Embryos and Ancestors. Clarendon Press, Oxford.

Verma. P.S and Agarwal, V.K. Chordate Embryology, S.Chand and Co. Ltd., New Delhi (1998).

Bodmer, Modern Embryology, Saunders International student edition, Philadelphia.3 rd Edition 1981.

Eli Benjamini et al.,(1991) Immunology – A short course – Wiley Publishers, NY.



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REFERENCES:

Hoar, W.S (1987) General and Comparative physiology, prentice – Hall.

M.K.Chanddrashekar – Circadian Rhythms – Madras science foundation, Chennai.

YEAR-II SEMESTER-III

PRACTICAL

(ANIMAL PHYSIOLOGY, AND DEVELOPMENTAL BIOLOGY)

I Major Practicals:

- 1) Qualitative analysis of digestive enzymes in cockroach.
- 2) Estimation of the rate of O_2 consumption in fish/crab with reference to body weight.
- 3) Detection nitrogenous waste products in fish tank water, bird excreta & mammalian urine.
- 4) Study of human salivary activity in relation to temperature.
- 5) Qualitative analysis of carbohydrates, proteins, and amino acids.

II Minor Practicals:

Kymograph –simple twitch, Trappe, Fatigue, Tetanus, Spigmanometer, pH meter, Colorimeter, Haemometer, Enzyme action – graphs (temperature, concentration of substrate and enzyme.)



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SCHEME

Semester – IV

No	Subject Code	Subject Type	Subject Title	Marks Allotted						
				Assignment Marks		Theory Marks		Practical Marks		Total Marks
				Max	Min	Max	Min	Max	Min	
1	FC-401/1	Core	English Language	10	4	40	14	-	-	50
2	FC-401/2	Core	Development of Entrepreneurship	10	4	40	14	-	-	50
3	BBT 141	Core	Molecular Immunology & Recombinant DNA Technology	20	8	80	27	50	17	150
4	BCH 141	Core	Based on Inorganic, Organic And Physical Chemistry	20	8	80	27	50	17	150
5	BSB 141	Optional	Structure, development & reproduction in flowering plants	20	8	80	27	50	17	150
	BSZ 141	Optional	Ecology & Evolutionary Biology	20	8	80	27	50	17	150
Total				80	32	320	109	150	51	550



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B.C.A./B.A./B.Com./B.Sc

Semester – IV

Course	Subject	Subject Code
B.Sc. (Biotechnology)	English Language	FC-401/1

Unit-I

1. William Wordsworth : “The World is Too Much With Us”
2. K. Aludiapillai : “Communication Education and Information Technology”
“Democratic Decentralisation”
3. S. C. Dubey : “Basic Quality of Life”
4. Sister Nivedita : “The Judgment Seat of Vikramaditya”
5. Juliun Huxley : “War as a Biological Phenomenon”
6. Robert Frost : “Stopping by Woods on a Snowy Evening”
7. Ruskin Bond : “The Cherry Tree”

Unit-II

Short Essay of about 250-300 words.

Unit-III

Translation of a short passage from Hindi to English.

Unit-IV

Drafting CV, writing e-mail message for official purpose.

Unit-V

Language Skills, One-word substitution, homonyms, homophones, words that confuse, Punctuation, Idioms



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B.C.A./B.A./B.Com./B.Sc

Semester – IV

Course	Subject	Subject Code
B.Sc. (Biotechnology)	Development of Entrepreneurship	FC-401/2

Unit-I

Entrepreneurship - Meaning, Concept, Characteristics of entrepreneur.

उद्यमिता – का आशय, मत, उद्यमिता के गुण।

Unit-II

Types of entrepreneurship, importance and views of various thinkers (Scholars).

- Formation of goals, How to achieve goals.
- Problems in achieving targets and solution.
- Self motivation, elements of self motivation and development.
- Views of various scholars, evaluation, solutions.

Leadership capacity : Its development and results.

उद्यमिता के प्रकार, महत्व और विभिन्न विद्वानों के मत, लक्ष्य निर्माण, लक्ष्य कैसे प्राप्त करें। लक्ष्य प्राप्ति में समस्याएँ, उनका समाधान, स्वप्रेरणा, स्वप्रेरणा के तत्व और विकास, विभिन्न विद्वानों के मत, आकलन, निष्कर्ष, नेतृत्व समता, उसका विकास और प्रतिफलन

Unit-III

Projects and various organisations (Govt., non-Govt.), Govt. Projects, Non-Govt. projects. Contribution of Books, their limitations, scope.

परियोजनाएँ तथा विभिन्न संगठन (शासकीय-अशासकीय), शासकीय परियोजनाएँ- अशासकीय परियोजनाएँ- बैंको का योगदान, उनकी सीमाएँ, क्षेत्र

Unit-IV

Functions, qualities, management of a good entrepreneur. Qualities of the entrepreneur (Modern and traditional). Management skills of the entrepreneur. Motive factors of the entrepreneur.

अच्छे उद्यमी के कौन-कौनसे कार्य, गुण, प्रबंधन इत्यादि, अच्छे उद्यमी के गुण आधुनिक और पूर्ववर्ती, उद्यमी की प्रबंधन कला, उद्यमी के प्रेरक तत्व

Unit-V

Problems and Scope of the Entrepreneur :

- Problem of Capital
- Problem of Power
- Problem of registration
- Administrative problems
- Problems of Ownership.

उद्यमी की समस्याएँ, क्षेत्र, पूँजी की समस्या, शक्तिकरण की समस्या, पूँजीवन की समस्या, प्रशासनिक समस्याएँ, स्वामित्व की समस्याएँ, इत्यादि



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Syllabus Semester – IV

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	MOLECULAR IMMUNOLOGY & RECOMBINANT DNA TECHNOLOGY	BBT-141

UNIT-I

Historical perspectives - overview of immune system, innate and acquired immunity, immune - systematic structure and organization.

Antigen and antigenicity, Immunoglobulins - structure, complements, antigen - antibody interaction - monoclonal antibodies.

UNIT-II

Organisation and expressions of immunoglobulin genes - histocompatibility complex.

Cytokines: Structure and function-cytokine receptors-biological functions of cytokines - cell mediated immunity; reports and T cell activation - Humoral response; B cell activation and proliferation - hypersensitive reactions.

UNIT-III

Immune regulation, autoimmunity, vaccines and immune response to infectious diseases - Immunodeficiency diseases (AIDS) - immune suppression & transplantation.

Unit -IV

History of Recombinant DNA Technology. Isolation and Quantification of DNA and RNA. Host controlled Restriction –Modification system, Restriction Endonucleases. Cutting and joining of DNA molecules *in vitro*. Phosphatases, Ligases and Polymerases. Vectors: Plasmid, Bacteriophage, Cosmids, Phagemid and other (SV 40) Virus vectors. Expression Vectors.

Unit -V

Gene transfer methods in animals: Microinjection, Electroporation, Microprojectile bombardment, Shot Gun method, Ultrasonication, Lipofection, Micro laser. Selection and Screening of Recombinants by genetic, immunochemical and hybridization methods, Southern, Northern and Western blotting.

REFERENCE

1. Immunology (V Edition),- Richard A.Goldsby, Thomas. J. Kindt, A. Osborne, JanisKuby, 2003. W.H. Freeman and company
2. Immunology, Ivan Roitt, 2001. Harcourt publishers, ltd.
3. Essential immunology, Ivan Riott, 2000. Blackwell Science, 9th Edition.
4. Immunology - An Introduction, Tizard.
5. Glick, B.R & Pasternak J.J (1994) Molecular Biotechnology, Principles and Applications of Recombinant DNA, American Society for Microbiology, Washington D.C
6. Christopler H. (1995) Gene cloning and Manipulating, Cambridge University Press



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7. Nicholl, D.S.T (1994) An Introduction of Genetic Engineering, Cambridge University Press.
8. Old. R.W. and Primrose, S.B. (1986) Principles of Gene manipulation, An introduction to genetic engineering (3rd Edition) Black well Scientific Publications
9. Watson J.D. Hopkins, N.H Roberts, J.W.Stectz J.A and Weiner A.M. (1988). Molecular biology of society for Microbiology
10. Lewin b. (1994) Genes VI, New York,Oxford University Press.

PRACTICE- IMMUNOLOGY

1. Blood grouping
2. Blood cell analysis
3. Lymphocyte subset identification and enumeration.
5. Routes of inoculation
6. Preparation of Antigen - Protocol of immunization
7. Methods of bleeding
8. Preparation of serum components
9. Immuno diffusion
11. Immuno electrophoresis
13. Haemoagglutination.
14. Serum electrophoresis.
15. Antigen-antibody reaction (precipitation & agglutination reaction tests)



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Syllabus Semester – IV

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	BASED ON INORGANIC, AND PHYSICAL CHEMISTRY	BCH-141

UNIT-I

Electronic structure, oxidation states and ionic radii and lanthanide contraction, complex formation, occurrence and isolation, lanthanide compounds. **Actinides**-General features and chemistry of actinides, chemistry of separation of Np, Pu and Am from U, Comparison of properties of Lanthanides and Actinides and with transition elements .

UNIT-II

Second law of thermodynamics, need for the law, , Carnot's cycles and its efficiency, Carnot's theorem, .Thermodynamics scale of temperature. Concept of entropy - entropy as a state function, entropy as a function of V & T, entropy as a function of P & T,

Thermodynamics-IV Third law of thermodynamics: Nernst heat theorem, statement of concept of residual entropy, evaluation of absolute entropy from heat capacity data. Gibbs and Helmholtz functions; Gibbs function(G) and Helmholtz function (A) as thermodynamic quantities, A & G as criteria for thermodynamic equilibrium and spontaneity, their advantage over entropy change. Variation of G and A with P, V and T.

UNIT-III

Electrolytic and Galvanic cells – reversible & Irreversible cells ,conventional representation of electrochemical cells. EMF of cell and its measurement, Weston standard cell, activity and activity coefficients .Calculation of thermodynamic quantities of cell reaction (ΔG , ΔH & K). Types of reversible electrodes – metal- metal ion gas electrode, metal –insoluble salt-anion and redox electrodes. Electrode reactions, Nernst equations, derivation of cell EMF and single electrode potential. Standard Hydrogen electrode, reference electrodes, standard electrodes potential, sign conventions electrochemical series and its applications.

UNIT-IV

Structure and nomenclature of amines, physical properties. Separation of a mixture of primary, secondary and tertiary amines. Preparation of alkyl and aryl amines (reduction of nitro compounds, nitriles, reductive amination of aldehydic and ketonic compounds. Gabrielphthalimide reaction, **Diazonium Salts** Mechanism of diazotisation, structure of benzene diazoniumchloride, Replacement of diazo group by H, OH, F, Cl, Br, I, NO₂reaction and its synthetic application.

Nitro Compounds Preparation of nitro alkanes and nitro arenes and their chemical reactions.



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UNIT-V

Nomenclature and structure of the carbonyl group. Synthesis of aldehydes and ketones with particular reference to the synthesis of aldehydes from acid chlorides, **Ketones** Mechanism of nucleophilic additions to carbonyl group with particular emphasis on benzoin, aldol, Perkin and Knoevenagel condensations. Condensation with ammonia and its derivatives. Wittig reaction. Mannich reaction. Oxidation of aldehydes, Baeyer–Villiger oxidation of ketones, Cannizzaro reaction. MPV, Clemmensen, Wolff-Kishner, LiAlH_4 and NaBH_4 reductions.

Book suggested :

1. I. L. Levine, Quantum Chemistry , Prentice-Hall Inc., New Jersey.
2. T. Engel and P. Reid, Physical Chemistry, Benjamin-Cummings.
3. S. M. Mukherji and S. P. Singh, Reaction Mechanism in Organic Chemistry, Macmillan
4. Advanced Inorganic Chemistry, F.A. Cotton and Wilkinson, John Wiley.

(Practicals)

SECTION – I (Inorganic)

Colorimetry:

1. To verify Beer - Lambert law for $\text{KMnO}_4/\text{K}_2\text{Cr}_2\text{O}_7$ and determine the concentration of the given $\text{KMnO}_4/\text{K}_2\text{Cr}_2\text{O}_7$ solution.
2. Preparations: Preparation of Cuprous chloride, prussian blue from iron fillings, tetraammine cupric sulphate, chrome alum, potassium trioxalatochromate (III).

Section-B (Physical)

3. To determine the enthalpy of solution of solid calcium chloride
- 4 .To study the distribution of iodine between water and CCl_4 .

Section-C (Organic)

Systematic identification (detection of extra elements, functional groups, determination of melting point or boiling point and preparation of at least one pure solid derivative) of the following simple mono and bifunctional organic compounds: benzoic acid, salicylic acid, aspirin, phthalic acid, cinnamic acid, benzamide, urea, acetanilide, benzanilide, aniline hydrochloride, p-toluidine, phenyl salicylate (salol), glucose, fructose, sucrose, *o*-, *m*-, *p*-nitroanilines, thiourea.

Books for practical:

1. Vogel's Textbook of Practical Organic Chemistry, A.R. Tatchell, John Wiley.
2. Practical Physical Chemistry, A.M. James and F.E. Prichard, Longman.
3. Findley's Practical Physical chemistry, B.P. Levitt, Longman.
4. Macroscale and Microscale Organic Experiments, K.L. Williamson, D.C. Heath.



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Fourth Semester

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	Structure, development & reproduction in flowering plants	BSB 141

UNIT –1

The Root system: Root apical meristems, Differentiation of primary and secondary tissues and their roles, Anatomy of Monocot and Dicot roots, Morphological modification of root for storage, respiration, reproduction and interaction with microbes.

UNIT – 2

The Shoot system: Shoot apical meristem and histological organization, Anatomy of primary stem in Monocotyledons and Dicotyledons, Secondary growth in stem and root – Vascular cambium and its functions, Characteristics of growth rings, Sapwood and Heart wood, Secondary Phloem, Cork Cambium and Periderm.

UNIT –3

The Leaf system: Origin, Development, Diversity in size, shape and arrangement, Internal structure of Dicot and Monocot leaf in relation to photosynthesis and water loss, Adaptations to water stress, senescence and abscission.

UNIT – 4

The Flower system: Concept of flower as a modified shoot, Structure of Anther, Microsporogenesis and Male Gametophyte, Structure of Pistil, Ovules, Megasporogenesis and Development of Female Gametophyte (Embryo Sac) and its types, Pollination – Mechanism and Agencies of Pollination, Pollen Pistil interactions and Self incompatibility.

UNIT – 5

Double Fertilization, Development and types of Endosperm and its morphological nature, Development of Embryo in Monocots and Dicots, Fruit development and maturation. Seed structure and dispersal, Vegetative Propagation.



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Practical Exercises + Scheme

(Marks- 50)

1- Cutting, staining and mounting of cross section of two materials of monocotyledons/dicotyledons root and stem and leaf like Sunflower and Maize or other available material. 15

2- Organisation of shoot Apex and Root Apex. 5

3- Study of Ovules and Anthers and their types 5

- Structure of stigma and style (Hibiscus, Maize, Ocimum, Citrus and Clitoria

(Aprajita) or plant studied by you.

4-Spotting- 10

5-Viva- voce- 5

6-Practical Record- 10

SUGGESTED READINGS:--

- 1•Gangulee, H.C., Das, K. S. And Dutta, C. 2007. College Botany Voll.I, New Central Book Agency (P) Ltd. Kolkata, 700009.
- 2•Hywood, V.H. & Moore, D.M. (eds) 1984. Current concepts in plant taxonomy. Acedemic press London.
- 3•Jones, S.B. Jr. and Luchsinger, A.E. 1986, Plant taxonomy (III edition) Mc Graw Hill Book Co. New York.
- 4•Maheshwari, P.1978. Plant Embryology.
- 5• Pandey, B. P. 2010. A Text book of Botany- Angiosperms, S. Chand & Company Ltd. Ramnagar, New Delhi- 110055.
- 6• Radford, A.E. 1986. Fundamentals of Plant Systematics, Harper and Row, New York.
- 7• Shrivastava and Das. Modern text book of Botany Vol-III & IV.
- 8• Singh, V., Pande P.C. and Jain , D. K. Structure & Development in Angiosperms. Rastogi Publication, Meerut.



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B.Sc.(Biotechnology)

Syllabus

Semester – IV

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	ECOLOGY & EVOLUTIONARY BIOLOGY	BSZ 141

UNIT I

Abiotic factors of the Environment : Temperature, Light and .Oxygen. Bio geo chemical cycles with special reference to Nitrogen Phosphorous and Carbon. Biotic factors of the environment - Animal relationship.

UNIT II

Population : characteristics –Natality, Mortality, Density, and age distribution, population control, life-tables, Food chains, Food webs and Ecological pyramids. Air pollution, Water pollution and Oil pollution. Noise pollution and Thermal pollution.

UNIT III

Pond as a Ecosystem, energy flow and ecological succession. Habitats – Terrestrial – Aquatic – Marine, Fresh water and estuary. Environmental resources- renewable and non renewable resources. Forest resources- Protection – Chipko movement- A forestation. Wild life management- Wild life sanctuaries and National Parks.

Unit-IV

History of Evolutionary thought - Origin of life –Chemical evolution. Evolution of self replicating systems –DNA world and RNA world.

Evidences from Paleontology -Comparative anatomy, Embryology, Physiology and Bio chemistry. Bio geography –Distribution in continents ,Continuous and discontinuous distribution - Endemism .

Unit-V

Natural selection- Species and Speciation- Sympatric and allopatric speciation. Isolating mechanism- mutation and genetic drift.

Adaptation and adaptive radiation, Colouration-mimicry-Darwins finches. Polymorphism-types and significance. Convergent-Divergent-parallel and co-evolution of Man and cultural evolution.

TEXT BOOKS

1. H.D.Kumar, Modern concepts of Ecology. Vikas Publishing house.
2. E.P. Odum, Fundamentals of Ecology.
3. G.C. Clarke, Elements of Ecology, John Wiley sons, New York
4. Rostogi, V.B. Organic Evolution, Kedernath, Ramnath publishers, Meerut.
- 5, Verma P.S. & Agarval, V.L. concepts of evolution S.Chand& Company.



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REFERENCES:

Introduction to evolution-Dodson-Evolution: process and product.

PRACTICAL

Spotters:

Description and uses of autoclave, Hot air oven, Incubator, Water bath, Centrifuge, Refrigerator, pH meter, Colori meter, Microtome, Rain gauge, Anemometer, Maximum minimum thermometer, Hygrometer, and Barometer.

Computer applications - Hardware of computer, storage device, mouse.

Submission of field Report.

Submission of Practical Record.



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SCHEME

Semester – V

No	Subject Code	Subject Type	Subject Title	Marks Allotted						
				Assignment Marks		Theory Marks		Practical Marks		Total Marks
				Max	Min	Max	Min	Max	Min	
1	FC-501/1	Core	Values & Spirituality	10	4	40	14	-	-	50
2	FC-501/2	Core	Basic Computer Information Technology-I	10	4	40	14	-	-	50
3	BBT 151	Core	Instrumentation & Basic Biostatistics	20	8	80	27	50	17	150
4	BCH 151	Core	Based On Inorganic, Organic And Physical Chemistry	20	8	80	27	50	17	150
5	BSB 151	Optional	Plant Physiology And Biochemistry	20	8	80	27	50	17	150
	BSZ 151	Optional	Microbiology And Biochemistry	20	8	80	27	50	17	150
Total				80	32	320	109	150	51	550



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B.C.A./B.A./ B.Com./B.Sc Semester – V

Course	Subject	Subject Code
B.Sc. (Biotechnology)	Values & Spirituality	FC-501/1

Chapter 6: THE SUPREME- SOURCE OF VALUES

6.1 Objectives	6.2 Introduction	6.3 All about the Supreme
6.4 Name & Form of the Supreme	6.5 Abode of the Supreme	
6.6 Attributes of the Supreme	6.7 Action of benediction	
6.8 Auspicious confluence (Now or never)		6.9 Summary
6.10 Glossary	6.11 Suggested reading	

Chapter 7 : HEALING RELATIONSHIPS WITH THE SUPREME

7.1 Objectives	7.2 Introduction	7.3 Significance of the relationships
7.4 All relations with ONE	7.5 major relations	7.6 True and eternal relations
7.7 Benefits of various relationships	7.8 The timeless dimensions	7.9 Summary
7.10 Glossary	7.11 Suggested reading	

Chapter 8 : RAJYOGA MEDITATION

8.1 Objectives	8.2 Introduction	8.3 Methods of Meditation
8.4 Rajyoga meditation with a difference	8.5 Five fold impact of Rajyoga meditation	
8.6 Stages of Rajyoga meditation	8.7 Attainments of Meditation	
8.8 Research studies on meditation	8.9 Summary	
8.10 Glossary	8.11 Suggested reading	

Chapter 9 : SPIRITUAL LIFE STYLE

9.1 Objectives	9.2 Introduction	9.3 Early morning meditation
9.4 Regular spiritual study	9.5 Authentic life style	9.6 Satwic Diet
9.7 Selfless service	9.8 Review of Personal Progress	
9.9 Summary	9.10 Glossary	9.11 Suggested reading

Chapter 10 : EXERCISES FOR PRACTICE

Quiet reflection- Practice introversion-Being an observer-Stand back and observe -Self awareness (Soul consciousness)-Experiencing Body free stage-Reflect on original qualities-Visualize the Divine-Think attributes of the Supreme-Developing a living relationship-Surrender to God-Create Good wishes for all-Visualization in Meditation: Orbs of Light- The forest-The Balloon



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B.C.A./B.A./ B.Com./B.Sc

Semester – V

Course	Subject	Subject Code
B.Sc. (Biotechnology)	Basic Computer Information Technology-I	FC-501/2

Unit – I

INTRODUCTION TO COMPUTER ORGANIZATION –I :- History of development of Computer system concepts. Characteristics, Capability and limitations. Generation of computer. Types of PC's Desktop, Laptop, Notebook. Workstation & their Characteristics.

कम्प्यूटर ऑर्गनाइजेशन का परिचय – I

कम्प्यूटर का इतिहास, कम्प्यूटर सिस्टम विचारधारा, विशेषताएं, योग्यता एवं सीमाएं, कम्प्यूटर की पीढ़ियां, पी.सी. के प्रकार, डेस्कटॉप के प्रकार, लेपटॉप के प्रकार, नोटबुक, वर्कस्टेशन आदि की विशेषताएं।

Unit – II

INTRODUCTION TO COMPUTER ORGANIZATION –II :- Basic components of a computer system Control Unit, ALU. Input/ Output function and Characteristics, memory RAM, ROM, EPROM, PROM.

कम्प्यूटर ऑर्गनाइजेशन का परिचय – II

कम्प्यूटर सिस्टम के आधार उपकरण, कंट्रोल युनिट, ए.एल.यू. इनपूट/ आउटपुट फंक्शन और विशेषताएं, मेमोरी रेम, रोम, ईपी रोम, पी रोम, और अन्य प्रकार की मेमोरी।

Unit – III

INPUT & OUTPUT DEVICES :- Input Devices : Keyboard, Mouse, Trackball. Joystick, Digitizing tablet, Scanners, Digital Camera, MICR, OCR, OMR, Bar-code Reader, Voice Recognition, Light pen, Touch Screen. **Output Devices:** Monitors Characteristics and types of monitor, Video Standard VGA, SVGA, XGA, LCD Screen etc. Printer, Daisy wheel, Dot Matrix, Inkjet, Laser, Line Printer. Plotter, Sound Card and Speakers.

इनपुट तथा आउटपुट डिवाइसेस

इनपुट डिवाइस : कीबोर्ड, माउस, ट्रैकबॉल, जॉयस्टिक, डिजिटाइजिंग टेबलेट, स्कैनर्स, डिजिटल केमरा, एमआईसीआर, ओसीआर, ओएमआर, बार कोड रीडर, आवाज को पहचानने वाला, लाइट पेन, टच स्क्रीन।

आउटपुट डिवाइस : मॉनीटर की विशेषताएं एवं मॉनीटर के प्रकार, वीडियो स्टैंडर्ड VGA, SVGA, XGA, LCD स्क्रीन आदि, प्रिंटर, डेजी व्हील, डॉट मैट्रिक्स, इंकजेट, लेजर, लाइन प्रिंटर, प्लॉटर, साउंड कार्ड्स एवं स्पीकर्स।

Unit – IV

STORAGE DEVICES :- Storage fundamental primary Vs Secondary. Various Storage Devices magnetic Tape. Cartridge Tape, Data Drives, Hard Drives, Floppy Disks, CD, VCD, CD-R, CD-RW, Zip Drive, DVD, DVD-RW.

स्टोरेज डिवाइजेस

स्टोरेज फंडामेंटल्स प्राइमरी विरुद्ध भिन्न स्टोरेज डिवाइजेस मेग्नेटिक टेप, कार्ट्रिज टेप, डाटा ड्राइव्स, हार्ड डिस्क ड्राइव्स, फ्लोपी डिस्कस, सी.डी. वी.सी.डी. सी.डी.-आर.सी.डी.- आर. डब्ल्यू, जीप ड्राइव, डी.वी.डी., डी.वी.डी., - आर. डब्ल्यू।



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Unit – V

INTRODUCTION TO OPERATING SYSTEM :- Introduction to operating systems, its functioning and types. basic commands of dos & Windows operating System.

Disk Operating System (DOS)

- Introduction, History and Versions of DOS.

DOS Basics

- Physical Structure of disk, Drive name, FAT, file & directory structure and naming rules, booting process, DOS system files.

DOS Commands

- Internal DIR, MD, CD, RD, Copy, DEL, REN, VOL, DATE, TIME, CLS, PATH, TYPE etc.
- External CHKDSK, SCOPE, PRINT DISKCOPY, DOSKEY, TREE, MOVE, LABEL, APPEND, FORMAT, SORT, FDISK, BACKUP, MODE, ATTRIB HELP, SYS etc.

ऑपरेटिंग सिस्टम का परिचय

ऑपरेटिंग सिस्टम का परिचय, उसके लक्षण एवं प्रकार, डॉस एवं विन्डोज का मूल कमांड। ऑपरेटिंग सिस्टम : डिस्क ऑपरेटिंग सिस्टम (DOS), परिचय इतिहास एवं वर्जन्स ऑफ डॉस।

डॉस बेसिक्स –

फिजीकल स्ट्रक्चर ऑफ डिस्क, ड्राइव नेम, फेट, फाईल एवं डायरेक्ट्री स्ट्रक्चर एवं नेमिंग नियम, बूटिंग प्रक्रिया, डॉस सिस्टम फाईल्स।

डॉस कमाण्डस–

- आंतरिक कमाण्डस DIR, MD, CD, RD, Copy, DEL, REN, VOL, DATE, TIME, CLS, PATH, TYPE आदि।
- बाह्य कमाण्डस CHKDSK, SCOPE, PRINT DISKCOPY, DOSKEY, TREE, MOVE, LABEL, APPEND, FORMAT, SORT, FDISK, BACKUP, MODE, ATTRIB HELP, SYS आदि।

Books Recommended-

- 1- डॉ. एस. के. विजय, डॉ. पंकज सिंह : कम्प्यूटर विज्ञान एवं सूचना प्रौद्योगिकी, मध्यप्रदेश हिन्दी ग्रन्थ अकादमी, भोपाल
- 2- डॉ. पंकज सिंह कम्प्यूटर अध्ययन, राम प्रसाद एड संस

Course	Subject	Subject Code
B.Sc. (Biotechnology)	Basic Computer Information Technology-I	FC 501/2

PRACTICALS

DOS :

- DOS commands : Internal & External Commands.
- Special batch file : Autoexec, Bar Hard disk setup.

Windows 98:

- Desktop setting : New folder, rename bin operation, briefcase, function. Control panel utility.
- Display properties: Screen saver, background settings.

Ms-Word:



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- Creating file: save, save as HTML, Save as Text, template, RTF Format.
- Page setup utility: Margin settings, paper size setting, paper source, layout.
- Editing: Cut, paste special, undo, redo, find, replace, goto etc.
- View file: page layout, Normal Outline, master document, ruler header, footer, footnote, full screen.
- Insert: break, page number, symbol, date & time, auto text, caption file, object, hyperlink, picture etc.
- Format: font, paragraph, bullets & numbering, border & shading, change case, columns.
- Table : Draw label, insert table, cell handling, table auto format, sort formula.



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Syllabus Semester – V

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	INSTRUMENTATION & BASIC BIOSTATISTICS	BBT-151

Unit -I

General Biophysical methods – Measurement of pH, Buffers, Henderson – Hasselbalch equation. Dielectric constant, Dipole moment, Osmosis, Diffusion, Isoelectric point.

Separation & Identification of Materials - concept of Chromatography (Partition Chromatography, Paper Chromatography, Adsorption Chromatography, TLC, GLC, Ion Exchange Chromatography, Gel Chromatography, HPLC, Affinity Chromatography) Electrophoresis (Gel Electrophoresis, Paper Electrophoresis).

Unit - II

Centrifugation – Basic Principle of Centrifugation, Instrumentation of Ultracentrifuge (Preparative, Analytical), Factors affecting Sedimentation velocity, Standard sedimentation Coefficient, Microscopy – Light microscopy, Bright & Dark Field microscopy, Fluorescence microscopy, Phase Contrast microscopy, TEM, SEM, X-Ray Crystallography, Xray diffraction, Bragg equation.

Unit - III

Spectroscopy: Simple theory of the absorption of light by molecules, Beer- Lambert law, Instrumentation for measuring the absorbance of visible light, Factors affecting the absorption properties of a Chromophore. Radioactive labeling & counting, Autoradiography. Scintillation counters, Geiger-Muller counter.

Unit - IV

Introduction and Definition of Biostatistics, Tabulation and classification of data, Frequency distribution and Graphical distribution of data, Measures of central tendencies Mean, Median, Mode and their properties. Computation of mean, variance and standard deviation, t-test, correlation coefficient.

Unit - V

Measures of dispersion: range, Mean deviation, Standard deviation and coefficient of Variation. Student T and Chi-square test, Concepts and problems on probability, Normal Distribution and their applications. Types of errors (Type I, II).

Suggested Readings

1. Biochemical Techniques theory and practice : White R
2. Analytical Chemistry: Christion G. D.
3. A Biologist Guide to Principle and Techniques: Willson K. and Gounding K. H.



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4. An Introduction to Practical Biochemistry: Plummer D. T.
5. Undergraduate Instrumental Analysis (1995) 5th ed., Robinsan, J. W.
6. Narayanan, P (2000) Essentials of Biophysics, New Age Int. Pub. New Delhi.
7. Roy R.N. (1999) A Text Book of Biophysics New Central Book Agency.
8. Bioinformatics(2002) Bishop Martin
9. Molecular databases for protein and sequence and structure studies: Sillince A. and Sillince M.
10. Sequence Analysis primers : Gribskov, M. and Devereux, J.
11. Bioinformatics: Sequence and Genome Analysis By David W.
12. Mount, *University of Arizona, Tucson*
13. Discovering Genomics, Proteomics, & Bioinformatics, Second
14. EditionBy A. Malcolm Campbell, *Davidson College*; Laurie J.
15. Heyer, *Davidson College*; With a Foreword by Francis S. Collins

PRACTICLE- INSTRUMENTATION

1. Determination of bacterial growth curve
2. Immobilization of enzymes using sodium alginate
3. Immobilization of yeast cells using sodium alginate
4. Ethanol production using immobilized yeast cells
5. Estimation of ethanol
6. Compound separation using column chromatography and thin layer chromatography
7. Fermentors

Introduction and working to all Instruments.



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Syllabus Semester – V

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	BASED ON INORGANIC, ORGANIC & PHYSICAL CHEMISTRY	BCH-151

UNIT-I

Limitations of valence bond theory, an elementary idea of crystal-field theory, crystal field splitting in octahedral, tetrahedral and square planar complexes, **Magnetic Properties of Transition Metal Complex** Types of magnetic behavior, methods of determining magnetic susceptibility, spin-only formula. L-S coupling, **Electron Spectra of Transition Metal Complexes** Types of electronic transitions, selection rules for d-d transitions, spectroscopic ground states, spectrochemical series. Orgel-energy level diagram for d1 and d9 states, discussion of the electronic spectrum of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ complex ion.

UNIT-II

Black-body radiation, Planck's radiation law, photoelectric effect, heat capacity of solids, Compton effect, wave function and its significance of Postulates of quantum mechanics, quantum mechanical operator, commutation relations, Hamiltonian operator, Hermitian operator, average value of square of Hermitian as apposite quantity, Role of operators in quantum mechanics, To show quantum mechanically that position and momentum cannot be predicated simultaneously, Determination of wave function & energy of a particle in one dimensional box, Pictorial representation and its significance.

UNIT-III

Introduction: Electromagnetic radiation, regions of spectrum, basic features of spectroscopy, **Rotational Spectrum** Diatomic molecules. Energy levels of rigid rotator (semi-classical principles), selection rules. **Vibrational spectrum** Infrared spectrum: Energy levels of simple harmonic oscillator, selection rules, pure vibrational spectrum, effects of an harmonic motion and isotopic effect on the spectra., idea of vibrational frequencies of different functional groups. **Raman Spectrum:** Concept of polarizability, pure rotational and pure vibrational Raman spectra of diatomic molecules, selection rules, Quantum theory of Raman spectra.

UNIT-IV

Classification and nomenclature. Monosaccharides, mechanism of osazone formation, inter conversion of glucose and fructose, chain lengthening and chain shortening of aldoses. Configuration of monosaccharides. Erythro and threo diastereomers. Conversion of glucose in to mannose. Formation of glycosides, ethers and esters. Determination of ring size of glucose and fructose. Open chain and cyclic structure of D(+)-glucose & D(-) fructose. Mechanism of mutarotation. Structures of ribose and deoxyribose. An introduction to disaccharides (maltose, sucrose and lactose) and polysaccharides (starch and cellulose) without involving structure determination.



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UNIT-V

Organomagnesium compounds: the Grignard reagents-formation, structure and chemical reactions. Organozinc compounds: formation and chemical reactions. Organolithium compounds: formation and chemical reactions.

Books:

1. P. S. Kalsi., Organic Reactions and their Mechanisms, New Age International
2. Chemistry of the Elements. N.N. Greenwood and A. Earnshaw, Pergamon.
3. Chemistry of the Elements. N.N. Greenwood and A. Earnshaw, Pergamon.
4. P.W. Atkins , Physical Chemistry, ELBS.

(Practical)

SECTION – I (Inorganic)

Semimicro qualitative analysis of mixture containing not more than four radicals (including interfering, Combinations and excluding insolubles):

Pb²⁺, Hg²⁺, Hg₂²⁺, Ag⁺, Bi³⁺, Cu²⁺, Cd²⁺, As³⁺, Sb³⁺, Sn²⁺, Fe³⁺, Cr³⁺, Al³⁺, Co²⁺, Ni²⁺, Mn²⁺, Zn²⁺, Ba²⁺, Sr²⁺, Ca²⁺, Mg²⁺, NH₄⁺, CO₃²⁻, S²⁻, SO₃²⁻, S₂O₃²⁻, NO₂⁻, CH₃COO⁻, Cl⁻, Br⁻, I⁻, NO₃⁻, SO₄²⁻, C₂O₄²⁻, PO₄³⁻, BO₃

Section-B (Physical)

1. To determine the strength of the given acid solution (mono and dibasic acid) conductometrically.
2. To determine the solubility and solubility product of a sparingly soluble electrolyte conductometrically
3. To determine the strength of given acid solution (mono and dibasic acid)/KMnO₄ – Mohr salt potentiometrically.

Section-C (Organic)

1. Laboratory Techniques

(a) **Steam distillation** (non evaluative) Naphthalene from its suspension in water Separation of o- and p-nitrophenols

(b) **Column chromatography** (non evaluative)

Separation of fluorescein and methylene blue Separation of leaf pigments from spinach leaves

2. Chromatography Method

Determination of R_f values and identification of organic compounds

(a) Separation of green leaf pigments (spinach leaves may be used) by paper chromatographic method

(b) Separation of a mixture of coloured organic compounds using common organic solvents by TLC.

3. Synthesis of the following organic compounds:

1. To prepare p-bromoaniline from p-bromoacetanilide.
2. To prepare m-nitroaniline from m-dinitrobenzene.



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Fifth Semester

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	Plant physiology and biochemistry	BSB 151

UNIT –1

Plant Water Relations: Properties of water, Importance of water in plant life, Diffusion, Osmosis & Osmotic relation to plant cell, Water Absorption, Ascent of Sap, Essential macro & micronutrients and their role. Transpiration: Structure & Physiology of Stomata, Mechanism of Transpiration, Factors affecting the rate of transpiration.

UNIT –2

Photosynthesis: Chloroplast, Photosynthetic pigments, Red drop, Emerson's effect, Concept of two Photosystems, Light reaction, Dark reaction - Calvin cycle, Hatch-Slack cycle, CAM cycle, Factors affecting rate of photosynthesis & Photorespiration.

UNIT –3

Respiration: Mitochondria, aerobic and anaerobic respiration, Respiratory coefficient, mechanism of respiration - Glycolysis, Kreb's cycle, Pentose phosphate pathway, Electron transport system, Factors affecting rate of respiration, Redox potential and theories of ATP synthesis.

UNIT – 4

Definition, classification and chemical structure: Monosaccharide, disaccharide, oligosaccharide and polysaccharides; Amino acids, essential and non essential amino acids; Lipids, saturated and non saturated fatty acids.

Classification, nomenclature and characteristics of Enzymes, Concept of holoenzyme, apoenzyme, co-enzyme and co-factors, mode & mechanism of enzyme action, Factors affecting enzyme activity. Plant Hormones, mode of action of Auxins, Gibberellins, Cytokinin and Abscissic acid.

UNIT – 5

Genetic Engineering: Tools and techniques of recombinant DNA technology; cloning vectors; genomic and cDNA library; transposable elements; gene mapping and chromosome walking. Biotechnology: Functional definition; basic aspects of plant tissue culture; cellular totipotency, differentiation and morphogenesis biology of Agrobacterium; vectors for gene delivery and marker genes; salient achievements in crop biotechnology.



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Practical Exercises + Scheme

(Marks- 50)

Question 1-

20

- 1- Preparation of solution of specific Normality, Molal and Molar solutions.
- 2- Exercises related to osmosis and osmotic relation.
- 3- Exercises related to Transpiration.
- 4- To separate Plastidial pigments by Paper Chromatography.
- 5- To perform the exercise of Photosynthesis & Respiration.
- 6- To perform biochemical test for Carbohydrate, Lipid and Protein.
- 7- To extract Enzyme for any plant part and demonstrate its activity.

(Any two experiments from above mentioned list)

Question 2: Comment on any technique related to Biotechnology-

05

Spotting-

10

Viva- voce-

5

Practical Record-

10

SUGGESTED READINGS:--

- 1• David, L. N. and Michael, M. C. 2000. Lehninger principle of Biochemistry, Macmillan worth Pub. New York, USA.
- 2• Gangulee, H.C., Das, K.S., Datta, C. and Sen, S. 2007. College Botany Voll.I, New Central Book Agency (P) Ltd. Kolkata, 700009.
- 3• Hopkins, W.G. 1995. Introduction of Plant physiology Pub. John wiley and sons New York.
- 4• Jain, V.K. 1974. Fundamentals of Plant Physiology, S. Chand & Compnay.
- 5• Pandey, B. P. 2010. A Text book of Botany- Angiosperms, S. Chand & Company Ltd. Ramnagar, New Delhi- 110055.
- 6• Taiz & Zeiger, E. 1998. Plant Physiology. Sinauer associates, Inc. Pub. Massachusetts U.S.A.
- 7• Verma, S.K. & Verma, M.A. 1995. Text book of Plant Physiology & Biotechnology. S. Chand & Company.
- 8• Verma, V. 1995. Plant Physiology, Emkey Pub.



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B.Sc.(Biotechnology)

Syllabus

Semester – V

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	MICROBIOLOGY AND BIOCHEMISTRY	BSZ 151

Unit-1

Classification of microorganisms- General characteristics of Bacteria, Virus, Yeast. Bacteria-Morphology, Bacterial cell structure, Motility, Nutrition and Reproduction. Virus-discovery-Morphology, Classification, phages and life cycle. Yeast-Morphology, cell structure, Multiplication, phages and cycle.

Unit-2

Morphology of water, air soil and sewage. Water-Microorganisms of water, total bacterial count. Air-Microorganisms in soil, nitrogen cycle. Sewage-Composition of sewage, treatment of sewage by microorganisms.

Unit-3

Food borne diseases- Microbial food poisoning by Salmonella and Clostridium botulinum (Botulism). Measures to prevent microbial food poisoning. Food infection-Food borne diseases- Diarrhea, Dysentery, Typhoid and Cholera. Water borne diseases-Hepatitis, Gastro enteritis, Campylobacter diarrhea, Giardia lamblia, Cryptosporidiosis cholera. Air borne diseases-Common cold, Tuberculosis, Pneumonia, Diphtheria.

Unit-4

Biochemistry- Definition and its importance, Physio- chemical forces acting on the living body –a) Definition of pH and its determination, Maintenance of pH of blood.
b) (Definition of osmosis, abnormality in edema and dehydration.
Nucleic acids, structure and classification.

Unit-5

Carbohydrates, lipids, Amino acids & proteins-Classification, structure and their function. Metabolism-Glycolysis-TCA cycle - Electron Transport chain, Urea cycle Deamination, Oxidation of fatty acids.

REFERENCES:

Microbiology- Pelzer.

Biology of Microorganism- Madigan-Brock

Microbiology Lab manual – Capachim.

Microbiology fundamentals and application- Atlas.R.M.

Principles of Biochemistry A.L. Lehninger, D.L. Nelson & M.M. Cox (1993) Worth publishes New York.

Biochemistry by L.Stryer (1994) freeman & co., Newyork.

Biochemistry by Zubay (1998) Macmillan publishers & co., New York.



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MAJOR PRACTICALS:

1. Estimation of dissolved oxygen content in the given water sample (Winklers method).
2. Estimation of salinity and pH in given water sample.
3. Plankton study –Identification and description of any five marine planktons.

MINOR PRACTICALS:

4. Examination of yeast, mould, protozoa and pathogenic bacteria.
5. Estimation of urine sugar.
6. Blood grouping.
7. Problems on calculation of Mean, median, mode.

Spotters

Developmental Biology-Slides

Slides of mammalian sperm and Ovum

Slides of different developmental stages of chick embryos (24, 48, 72, 96 hrs)

Slides of blastula and gastrula of frog (morula, early gastrula, yolk plug stage, late gastrula)

Placenta of Sheep / Pig/ Rat.



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SCHEME

Semester – VI

No	Subject Code	Subject Type	Subject Title	Marks Allotted						
				Assignment Marks		Theory Marks		Practical Marks		Total Marks
				Max	Min	Max	Min	Max	Min	
1	FC-601/1	Core	Values & Spirituality	10	4	40	14	-	-	50
2	FC-601/2	Core	Basic Computer Information Technology-I	10	4	40	14	-	-	50
3	BBT 161	Core	Bioinformatics & Tissue Culture Technology	20	8	80	27	50	17	150
4	BCH 161	Core	Based On Inorganic, Organic And Physical Chemistry	20	8	80	27	50	17	150
5	BSB 161	Optional	Plant Ecology, Biodiversity And Phytogeography	20	8	80	27	50	17	150
	BSZ 161	Optional	Biostatistics And Computer Applications & Basic Animal Biotechnology	20	8	80	27	50	17	150
Total				80	32	320	109	150	51	550



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B.C.A./B.A./B.Com./B.Sc Semester – VI

Course	Subject	Subject Code
B.Sc. (Biotechnology)	भाषा कौशल एवं व्यक्तित्व विकास	FC-601/1

इकाई –1

- | | |
|--|-------------------------|
| 1. भारतीय संस्कृति | 2. भारतीय समाज व्यवस्था |
| 3. सम्यता एवं संस्कार | 4. वैश्विक चेतना |
| 5. समन्वयीकरण (भारतीय एवं अंतर्राष्ट्रीय संदर्भ में) | |

bdkb & 2

- | | | |
|---------|------------|----------|
| 1. धर्म | 2. न्याय | 3. दर्शन |
| 4. नीति | 5. साहित्य | |

bdkb & 3

- | | |
|---|----------------|
| 1. संचार संसाधन : सम्पर्क के नए क्षितिज | 2. समाचार पत्र |
| 3. भारतीय प्रेस परिषद् | 4. रेडियों |
| 5. दूरदर्शन | |

bdkb & 4

- | | | |
|--------------------------------|--------------|----------|
| 1. सिनेमा | 2. रंगमंच | 3. संगीत |
| 4. चित्र, मूर्ति, स्थापत्य कला | 5. शिल्प कला | |

bdkb & 5

- | | |
|--|---------------------------------------|
| 1. कम्प्यूटर | 2. दूरभाष : विज्ञान की सौगात |
| 3. मंत्र (कहानी) : प्रेमचंद | 4. मातृभूमि (कविता) : मैथिलीशरण गुप्त |
| 6. साहित्यकार का दायित्व : डॉ. प्रेम भारती | |

1. nHk iLrd & e/; i.ni'k fgUnh xifk vdkneh] Hkkiky }kjk i:dkf'kr iLrd



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B.Sc.(Biotechnology)

B.C.A./B.A./B.Com./B.Sc

Semester – VI

Course	Subject	Subject Code
B.Sc. (Biotechnology)	Basic Computer Information Technology-II	FC-601/2

Unit-I

Word Processing: Word

- Introduction to word Processing.
- MS Word: features, Creating, Saving and Operating Multi document windows, Editing Text selecting, Inserting, deleting moving text.
- Previewing documents, Printing document to file page. Reduce the number of pages by one.
- Formatting Documents: paragraph formats, aligning Text and Paragraph, Borders and shading, Headers and Footers, Multiple Columns.

Unit-II

Introduction to Excel

Excel & Worksheet:

- Worksheet basic.
- Creating worksheet, entering data into worksheet, heading information, data text, dates, alphanumeric, values, saving & quitting worksheet.
- Opening and moving around in an existing worksheet.
- Toolbars and Menus, keyboard shortcuts.
- Working with single and multiple workbook coping, renaming, moving, adding and deleting. coping entries and moving between workbooks.
- Working with formulas & cell referencing.
- Autosum.
- Coping formulas
- Absolute & Relative addressing.

Unit-III

Introduction to Power Point

- Features and various versions.
- Creating presentation using Slide master and template in various colour scheme.
- Working with slides make new slide move, copy, delete, duplicate, lay outing of slide, zoom in or out of a slide.
- Editing and formatting text: Alignment, editing, inserting, deleting, selecting, formatting of text, find and replace text.

Unit-IV



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Power Point– II

- Bullets , footer, paragraph formatting, spell checking.
- Printing presentation Print slides, notes, handouts and outlines.
- Inserting objects Drawing and Inserting objects using Clip Arts picture and charts.
- Slide sorter, slide transition effect and animation effects. Presenting the show making stand alone presentation, Pack and go wizards.

Unit – V

Evolution, Protocol, concept, Internet, Dial-up connectivity, leased line, VSAT, Broad band, URLs, Domain names, Portals. E-mail, Pop & web based Email. Basic of sending and receiving Emails, Email & Internet Ethics, Computer virus, Antivirus software wage, Web Browsers.

Books Recommended-

1. डॉ. एस. के. विजय, डॉ. पंकज सिंह : कम्प्यूटर विज्ञान एवं सूचना प्रौद्योगिकी, मध्यप्रदेश हिन्दी ग्रन्थ अकादमी, भोपाल
2. डॉ. पंकज सिंह कम्प्यूटर अध्ययन, राम प्रसाद एड संस

Course	Subject	Subject Code
B.Sc. (Biotechnology)	Basic Computer Information Technology-I	FC 601/2

Practical

Ms-Power Point:

Creating new slide, formatting slide layout, slide show & sorter, Inserting new slide, slide no., date, time, chart, formatting slide, tool operation.

List of suggested practical work:

- Under standing of a dial up connection through modern.
- Configuring a computer for an e-mail and using outlook Express or Netscape Messenger.
- Registration an e-mail address.
- Understanding of e-mail drafting.
- Understanding of address book maintenance for e-mail.
- Understanding of different mail program tools.
- Send and receive functions of e-mail.



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B.Sc.(Biotechnology)

Syllabus Semester – VI

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	BIOINFORMATICS & TISSUE CULTURE TECHNOLOGY	BBT-161

UNIT-I

Bioinformatics-definition, history; computer - system, topology and peripherals for communication; Internet - basics, connection, web browsing and URL

Data bases - Nucleic acid sequence data bases (NCBI, EMBL, DDJB), Protein sequence data base-SWISS-PORT, data base searching - BLAST.

UNIT-II

Introduction to Techniques - Introductory history, Laboratory organization, Maintaining Aseptic environment, Basic concepts in cell culture - cell culture, Cellular Totipotency, Somatic Embryogenesis. In vitro culture : approaches & methodologies - preparation steps for tissue culture, surface sterilization of plant tissue material, basic procedure for aseptic tissue transfer, incubation of culture.

UNIT-III

Tissue nutrition : Growth Hormones - Plant cells (Composition of culture media, Growth hormones, Vitamins, Unidentified supplements, selection of media); Animal cells (substrate on which cells grow, Feeder layer on substrate, gas phase for tissue culture, media and supplements).

UNIT-IV

Tissue culture methodologies - Plant cells (Callus Culture, Cell Suspension Culture, Organ culture); Animal cells (Source of tissue, primary culture, differentiation of cells, growth kinetics, animal cell lines and their origin and characterization).

UNIT-V

Cloning & Selection of specific cell types – cloning, somatic cell fusion and HAT election, Medium suspension fusion, selection of Hybrid clone, production of monoclonal antibodies, Organ Culture - Culture of embryonic organs, whole embryo culture, culture of adult organs.

Reference

1. Programming in ANSIC, E. Balagurusamy, 1991. Tata Mcgraw Hill.
2. Introduction to bioinformatics, 2001. AH wood, T.K. Parry smith DJ, Pearson education Asia.
3. C & Unix programming; A conceptual perspective, 1995. Kulti, Tata Mc Graw Hill.



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B.Sc.(Biotechnology)

4. Developing bioinformatics in computer skill, Gibas C, Jambeek P.S, oreilly, 2001. associates inc. Shroff publishes.
5. Plant tissue culture : Bajaj, Y.P.S. Series.
6. Plant tissue culture : Gamborg and Phillip.
7. Basic and Agricultural Biotechnology (1993) Purohit and Mathur
8. Plants, Genes, and Agriculture : Chrispeels, M. J. and Grierson, D.
9. Genetic Engineering of crop plants : Lycett, G. W. and Grierson, D.
10. Biotechnological innovation in Animal productivity: (Biotol Series)
11. Culture of Animal cell: A mannual of Basic Techniques(4th ed.) (2000) Freshney

PRACTICE:

1. Primary cell culture
2. Continuous cell culture
3. Drug/Toxicity testing
4. Assessment of genetic variation related to plant taxa using allozyme method
5. Tissue culture methods-media preparation, sterilization, inoculation of explants, callus culture, suspension cultures, anther and ovule cultures.
6. Isolation of protoplasts, viability test for protoplasts, protoplast culture.
7. Working gel documentation system and analysis of electrophoretic gels.
8. Quantification of DNA and RNA in plant tissues by spectrophotometer method.
9. Quadratic equations
10. Mean, standard deviation
11. Factorial (using subroutine)
12. NCR using subroutine
14. Prime numbers
15. Largest and smallest numbers



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B.Sc.(Biotechnology)

Syllabus Semester – VI

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	Based on inorganic, Organic & Physical Chemistry	BCH-161

UNIT-I

Arrhenius, Bronsted – Lowry, the Lux – Flood, Solvent system and Lewis concepts of acids & bases, relative strength of acids & bases, Concept of Hard and Soft Acids & Bases. Symbiosis, electronegativity and hardness and softness.

Bioinorganic Chemistry

Essential and trace elements in biological processes, metalloporphyrins with special reference to haemoglobin and myoglobin. Biological role of alkali and alkaline earth metal ions with special reference to Ca^{2+} . Nitrogen fixation.

UNIT-II

Interaction of radiation with matter, difference between thermal and photochemical processes. Laws of photochemistry: Grotthus-Draper law, Stark-Einstein law (law of photochemical equivalence) Jablonski diagram depicting various processes occurring in the excited state, qualitative description of fluorescence, phosphorescence, non-radiative processes (internal conversion, intersystem crossing), quantum yield, photosensitized reactions-energy transfer processes (simple examples).

UNIT-III

Statement and meaning of the terms – phase component and degree of freedom, thermodynamic derivation of Gibbs phase rule, phase equilibria of one component system – Example – water and Sulphur systems. Phase equilibria of two component systems solid-liquid equilibria, simple eutectic Example Pb-Ag system, desilverisation of lead.

UNIT-IV

Introduction: Molecular orbital picture and aromatic characteristics of pyrrole, furan, thiophene and pyridine. Methods of synthesis and chemical reactions with particular emphasis on the mechanism of electrophilic substitution. Mechanism of nucleophilic substitution reactions in pyridine derivatives. **Heterocyclic Compounds-II** Introduction to condensed five and six- membered heterocyclic. Preparation and reactions of indole, quinoline and isoquinoline with special reference to Fisher indole synthesis, Skraup synthesis and Bischler-Napieralski synthesis.



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UNIT-V

Nomenclature, structural features, Methods of formation and chemical reactions of thiols, thioethers, sulphonic acids, sulphonamides and sulphaguanidine. Synthetic detergents alkyl and aryl sulphonates. **Amino Acids, Peptides & Proteins**

Classification, of amino acids. Acid-base behavior, isoelectric point and electrophoresis. Preparation of β -amino acids. Structure and nomenclature of peptides and proteins. Classification of proteins. Peptide structure determination, end group analysis, selective hydrolysis of peptides. Classical peptide synthesis, solid-phase peptide synthesis. Structures of peptides and proteins: Primary & Secondary structure.

Books:

1. Ira N. Levine, Quantum Chemistry, Prentice Hall.
2. F. A. Carey and R. J Sundberg, Advanced Organic Chemistry, Part A and B, Plenum
3. K.J. Laidler, Chemical Kinetics. McGraw-Hill.
4. Inorganic Chemistry, J.E. Huhey, Harpes & Row.
5. Valence, C. A. Coulson, Oxford University Press.

(Practical)

Section-B (Physical)

1. To determine the molecular weight of a non-volatile solute by Rast method.
2. To standardize the given acid solution (mono and dibasic acid) pH metrically.

Section-C (Organic)

2. Chromatography Method

Determination of R_f values and identification of organic compounds

(a) Separation of green leaf pigments (spinach leaves may be used) by paper chromatographic method

(b) Separation of a mixture of coloured organic compounds using common organic solvents by TLC.

3. Synthesis of the following organic compounds:

1. To prepare o-chlorobenzoic acid from anthranilic acid.
2. To prepare S-Benzyl-iso-thiourea from thiourea.

Book for practical:

1. Synthesis and Characterization of Inorganic Compounds, W.L. Jolly. Prentice Hall.
2. Experiments and Techniques in Organic Chemistry, D.P. Pasto, C. Johnson and M. Miller, Prentice Hall.
3. Practical Physical Chemistry, A.M. James and F.E. Prichard, Longman.



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B.Sc.(Biotechnology)

Sixth Semester

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	Plant ecology, biodiversity and phytogeography	BSB 161

UNIT – 1

Ecosystems: Structure and types, Biotic and Abiotic components, Trophic levels, Food chains, Food webs, Ecological pyramids, Energy flow; Biogeochemical cycles: Concept, Gaseous and Sedimentary cycles, Carbon, Nitrogen, Phosphorus and Sulphur cycle.

UNIT – 2

Ecological adaptations: Morphological, Anatomical and Physiological responses, Water adaptation (Hydrophytes, Xerophytes and Mesophytes), Temperature adaptation (Thermoperiodism and Vernalization), Light adaptation (Heliophytes and Sciophytes), Plant Succession: Causes, trends and processes, types of succession - Lithosere, Hydrosere and Xerosere.

UNIT – 3

Population Ecology: Distribution patterns, Density, Natality, Mortality, Growth curves, Ecotypes and Ecads; Community Ecology: Characteristics, Classification, Life forms.

Biodiversity: Basic concept, definition, Importance, Biodiversity of India, Hotspots, In situ and ex situ conservation, Endangered and threatened species, Red data book.

UNIT – 4

Soil: Physico-chemical properties, Soil formation, Development of Soil Profile, Soil classification, Soil composition, Soil factors; Pollution: Definition, Types & Causes; Global warming, Climate change and Ozone holes.

UNIT – 5

Phytogeography: Phytogeographical regions of India, Vegetation -types of Madhya Pradesh, Biosphere reserves, Sanctuaries and National parks of Madhya Pradesh, Natural resources – definition and classification of natural resources, Conservation and management of natural resources, Land resources management, Water resources management, Wet land resource management.



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Practical Exercises + Scheme

(Marks- 50)

1-To determine the minimum size of Quadrat by species area curve method.	05
- To conduct exercise on Frequency, Density and Abundance.	
2- Study of soil with reference to soil texture, water holding capacity, pH and test for Carbonate and Nitrate.	05
3-Preparation of slides of Xerophytic, Hydrophytic and Mesophytic plants.	10
4-To comment upon Phytogeographic region (model/ charts) and National Parks (Photographs).	05
5-Spotting-	10
6-Viva- voce-	5
7-Practical Record-	10

SUGGESTED READINGS:--

- 1• Banerjee, S.1998. Bio diversity conservation- Agrobotamica, Bikaner.
- 2• Kumar, U.K 2006. Bio diversity principles and conservation, Agrobios, Jodhpur.
- 3• Odum, E.P. 5Th ed. 2004 .Fundamentals of Ecology. Natraj Publisher, Dehradun.
- 4• Puri, G.S. 1960. Indian Forest Ecology.
- 5• Sharma, P.D. 7th ed. 1998.Ecology and Environment, Rastogi Publication, Shivaji Road. Meerut, 250002. India.
- 6• Shukla, R. S. & Chandel, P.S. 2006. A Text book of Plant Ecology.



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B.Sc.(Biotechnology)

Syllabus Semester – VI

BRANCH	SUBJECT TITLE	SUBJECT CODE
B.Sc. (Biotechnology)	BIostatISTICS AND COMPUTER APPLICATIONS & BASIC ANIMAL BIOTECHNOLOGY	BSZ 161

Unit-1

Introduction-definition, data types – primary and secondary – Classification of data, Collection of data – tabular and graphical representation – Bar diagram, Pi diagram, Column graph, Histogram, Ogive curves.

Unit-2

Measures of central tendency – Mean, Mode and Median, Variance, Standard deviation, Standard error and Coefficient of variance. Simple Correlation, Simple Regression, Chi square test, student's – t- test.

Unit-3

Fundamentals of Computer: Classification, Computer organization, Input devices, processing unit, output devices, external storage devices, software, WWW, CONCEPT OF E-Mail.

Computer and its application to biology-Definition and scope of Bioinformatics - application and introduction to Biological data.

Unit-4

History of animal cell culture, Laboratory requirements for animal cell culture, Sterilization techniques. Media used for animal cell culture, Types of cell culture (Primary and Secondary), Introduction to established cell lines, Stem cells.

Unit-5

Cryopreservation of cell cultures, application of animal cell cultures in production of therapeutics proteins. Hybridoma technology.

Gene transfer methods in Animals – Microinjection, Embryonic Stem cell gene transfer, Retrovirus & Gene transfer. Transgenic Animals, Animal propagation – Artificial insemination,

REFERENCES:

Introduction of Biostatistics and Computer Science – Y.I Parkar & M.G Dhanyagude NiraliPrakashan publishers, Pune.

Biostatistics by K.S. Negi ATIBS publications & distributors, New Delhi.

Bishop O.N. Statistics for Biology. Boston, Hollghtan, Mifflin.

Introduction to Biostatistics by Pranab kumar, S.Chand company Ltd. New Delhi.